LSI-lite Case Studies (with Color Telemetry)
A profiled composition stress test across Δx, r_v, ρ_r
Where images hold or break

What this is (and isn't)
LSI-lite is a **structural** read, not a taste meter. It runs a small, profiled composition test over three gauges and reports how far an image sits from healthy bands for its profile.

- **Δx — off-center gravity:** how decisively the subject sits away from geometric center.
- **r_v — void ratio:** how much *true* empty space remains (via a non-semantic foreground mask).
- **ρ_r — rupture energy:** local surface/edge pressure (Laplacian energy in a subject+halo region).

New: color telemetry (non-gating)
Alongside the standard gray/Otsu mask, we log a color-informed foreground mask (LAB k-means). This adds two advisory fields:

- `rᵥ_color_mask` — void ratio computed from the color mask.
- `dx_color_L` — Δx recomputed on the L-channel with that mask.
  Color **does not change acceptance**; it's an **audit** that explains divergences in tonal or reflective scenes.

Profiles, bands, and the gate
Each set is read under a **profile** (Figure_Default, MarkMaking_Expressive, Landscape), which carries **band guards** and **weights** for the three primitives.
An image is **inside the gate** only if **LSI_lite_100 ≥ 55** *and* all three bands are **OK** (no RED). Two failure regimes matter:

1. **Band violation** (any RED), regardless of composite score.
2. **Under-gate** (all bands OK, but composite < 55).

Think of the gate as a **safety zone**, not an aesthetic verdict. We use it to find **deviants** (interesting outliers) and to name the knob that's pulling them out.

What you'll see for each set

- **Signals:** Which gauges and profile are in play, plus the gate rule.
- **Results (run facts):** The row card only—frames, band flags, LSI_lite_100, and the **priority knob** (which primitive is farthest from center).
- **Directions-to-center:** Whether Δx, r_v, or ρ_r needs to increase/decrease to move toward band center.
- **Researcher's Take:** Method-minded interpretation of the telemetry.
- **Artist's Take:** Composition-minded read on placement, space, and mark energy.
- **Observations (LSI reality):** Why the gate behaved as it did; profile effects.
- **Plain English:** A short, non-specialist summary.
- **Did the LSI Hold Up?:** Whether behavior matched the stated intent.
- **Caveats (expected, not bugs):** Non-semantic masks, high-frequency sensitivity, profile bias, etc.
- **What I'd Do Next:** Concrete nudges tied to the priority knob.
- **Color check (telemetry only):** Brief note when color and gray disagree (and why).

Reading the numbers quickly

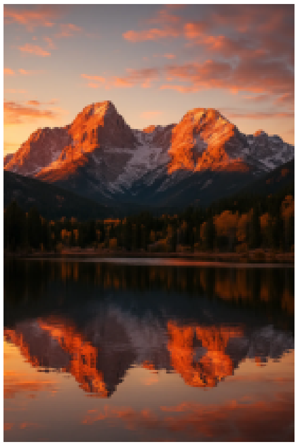
- **Bands** tell you if a primitive is within its healthy range; **RED** is a hard stop.
- **LSI_lite_100** is a weighted, band-aware composite (geometric mean) on a 0–100 scale.
- **Priority knob / score** rank which primitive is pulling the image farthest from its band center—use this to fix or to *intentionally* push the deviant.

Limitations to remember
Masks are **non-semantic** (they don't know "person/chair/window"), high-frequency debris can peg ρ_r, and profiles encode **taste-free** structural priors (e.g., Landscape weights r_v higher). That's by design for a portable, proof-of-concept read.

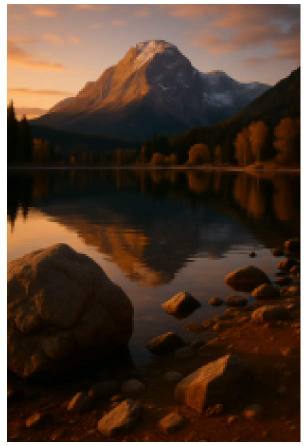
Case Studies

1. Landscape

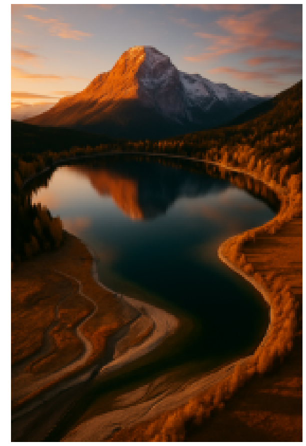
Landscape0.png



Landscape1.png

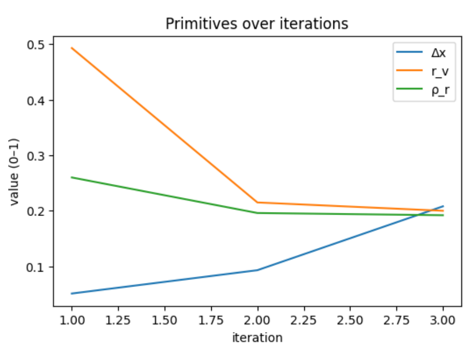


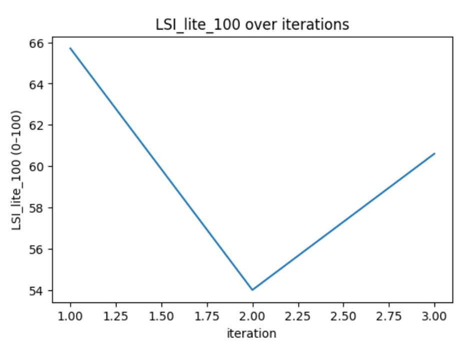
Landscape2.png



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.051	0.493	0.26	0.657	65.7	OK	OK	OK	TRUE	Landscape0.png	Landscape	TRUE	0.449	0.998	increase
1	0.093	0.215	0.196	0.54	54	OK	OK	OK	FALSE	Landscape1.png	Landscape	TRUE	0.407	0.904	increase
2	0.208	0.2	0.192	0.606	60.6	OK	OK	OK	TRUE	Landscape2.png	Landscape	TRUE	0.292	0.649	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.007	0.018	increase	TRUE	0.18	0.5	increase	dx	0.349	0.644	0.101	color	LAB	ok
TRUE	0.285	0.712	increase	TRUE	0.244	0.678	increase	dx	0.317	0.789	0.125	color	LAB	ok
TRUE	0.3	0.75	increase	TRUE	0.248	0.689	increase	rv	0.3	0.777	0.204	color	LAB	ok





Signals: Δx, r_v, ρ_r • **Gate:** LSI_lite_100 ≥ 55 **and** no RED

Results (run facts, trimmed)

- **Rows (0→2)**
  - **0)** Δx **0.051** | r_v **0.493** | ρ_r **0.260** | **65.7** | **ACCEPT** | bands **OK** | **knob:** Δx **0.349**
  - **1)** Δx **0.093** | r_v **0.215** | ρ_r **0.196** | **54.x** | **FAIL** (below gate) | bands **OK** | **knob:** Δx **0.317**
  - **2)** Δx **0.208** | r_v **0.200** | ρ_r **0.192** | **60.6** | **ACCEPT** | bands **OK** | **knob:** r_v **0.300**

Directions-to-center

- **Δx: increase** on all three (norm ≈ **0.998** → **0.904** → **0.649**).
- **r_v: increase** on all three (norm ≈ **0.018** → **0.712** → **0.750**).
- **ρ_r: increase** on all three (norm ≈ **0.319** → **0.542** → **0.556**).

Researcher's Take

- The **dip at #1** is driven by **r_v contraction** (Landscape weights r_v highest at 0.40). Δx improves, but the score drops because the image reads more filled-in.
- **Recovery at #2** comes from Δx continuing to improve while r_v rises back within band; ρ_r stays steady.
- **Gate precedence:** #1 **fails** only because the composite falls just under 55; **all bands are OK** in the series.

Artist's Take

- **#0:** Clean off-center mountain with ample sky/water; clear breathing room.
- **#1:** Reflection mass crowds the center; planes meet awkwardly; frame feels tight.
- **#2:** Keeps the off-center anchor and opens the scene; score climbs, but a touch more open sky/water would help.

Observations (LSI reality)

- Acceptance in this profile is often **r_v-limited**; here the scores track r_v, not Δx.
- No RED bands anywhere; tool behavior is a **within-band rebalance**, not a gate trip.

Plain English

- First and third pass; the middle one **fails narrowly** because it feels **too filled**.

Did the LSI Hold Up?

- **Yes.** It reflected the Landscape trade-off: off-center got better, but the set only "reads" as open when r_v doesn't collapse.

Caveats (expected, not bugs)

- **Profile bias:** r_v gets the **highest weight** here; Δx gains alone won't raise the score if breathing room shrinks.
- Reflection scenes can make Δx look "healthy" while r_v penalizes a filled frame—by design.

What I'd Do Next

- **Hold Δx gains** (keep the mountain clearly off-center).
- **Open r_v:** add more sky/water or a cleaner vent; avoid over-polish.
- Add a **bit of surface pressure** so the image doesn't feel overly smooth.

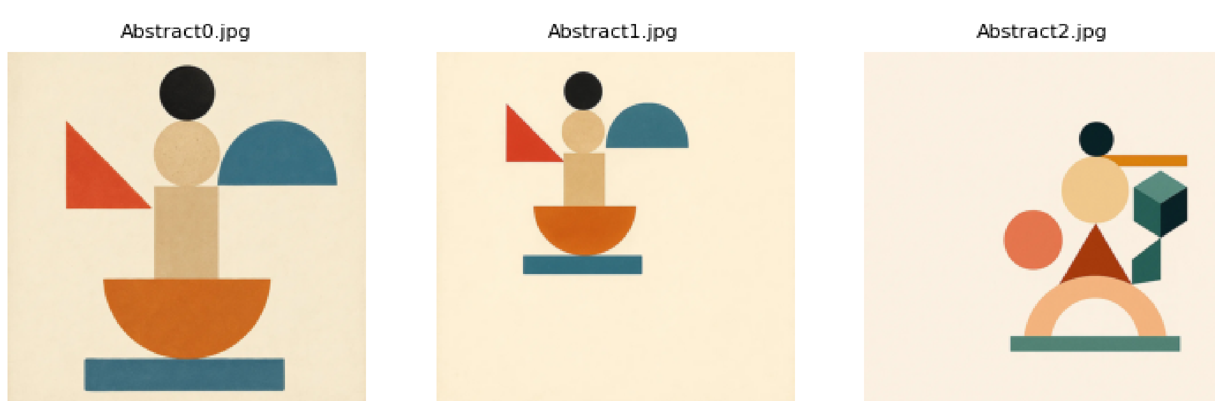
Color check (telemetry only)

- **r_v_color_mask vs gray r_v: 0.644 vs 0.493**, **0.789 vs 0.215**, **0.777 vs 0.200** → color sees **more void** (separates sky/water from land better than gray Otsu).
- **Δx on L with the color mask: 0.101 / 0.125 / 0.204** vs gray Δx **0.051 / 0.093 / 0.208** (subject-only centroid reads slightly farther off-center at #0/#1).
- **Takeaway:** In warm reflection/tonal scenes, gray **underestimates "breathing"**; color telemetry clarifies the subject/ground split. (Telemetry only; acceptance unchanged.)

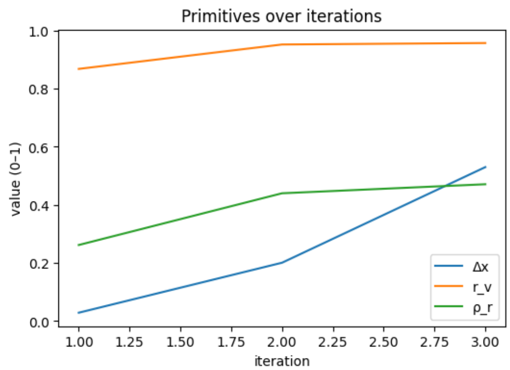
Color audit

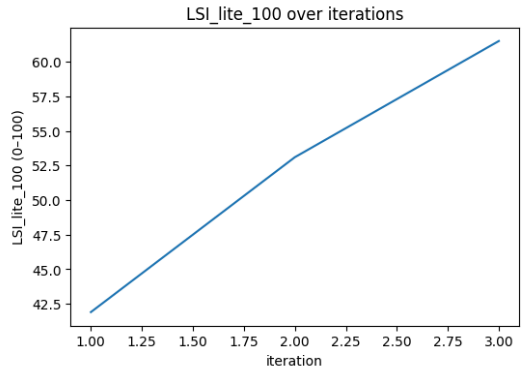
- r_v_color_mask vs gray r_v → **0.644 vs 0.493**, **0.789 vs 0.215**, **0.777 vs 0.200** (+0.151 / +0.574 / +0.577).
- Δx on L with color mask → **0.101 / 0.125 / 0.204** vs Δx **0.051 / 0.093 / 0.208** (small–moderate).
- **Takeaway:** Reflections/tonals: **void↑ in color**; gate still tracks r_v (gray) and Δx.

2. Abstract



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.029	0.868	0.262	0.419	41.9	RED	OK	OK	FALSE	Abstract0.jpg	Figure_Default	FALSE	0.421	1	increase
1	0.201	0.952	0.44	0.531	53.1	OK	RED	OK	FALSE	Abstract1.jpg	Figure_Default	TRUE	0.249	0.622	increase
2	0.53	0.957	0.471	0.615	61.5	OK	RED	OK	FALSE	Abstract2.jpg	Figure_Default	TRUE	0.08	0.2	decrease
rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status	
TRUE	0.368	0.92	decrease	TRUE	0.188	0.537	increase	dx	0.45	0.869	0.032	color	LAB	ok	
FALSE	0.452	1	decrease	TRUE	0.01	0.029	increase	rv	0.35	0.952	0.19	color	LAB	ok	
FALSE	0.457	1	decrease	TRUE	0.021	0.06	decrease	rv	0.35	0.865	0.407	color	LAB	ok	





Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands

Results

Rows (trimmed):

- **0)** Δx **0.029** | r_v **0.868** | ρ_r **0.282** | **41.9** | **REJECT** | knob **dx** (0.45) | bands: **Δx RED**, r_v OK, ρ_r OK | r_v_color_mask **0.869** | dx_color_L **0.032**
- **1)** Δx **0.291** | r_v **0.962** | ρ_r **0.440** | **53.1** | **REJECT** | knob **r_v** (0.35) | bands: **r_v RED**, others OK | r_v_color_mask **0.952** | dx_color_L **0.190**
- **2)** Δx **0.083** | r_v **0.953** | ρ_r **0.471** | **61.5** | **REJECT** | knob **r_v** (0.35) | bands: **r_v RED**, others OK | r_v_color_mask **0.865** | dx_color_L **0.467**

Researcher's Take

- Score trend climbs (≈42→62) on **Δx↑** and **ρ_r↑**, but **gate fails** because #1–#2 push r_v above the profile's upper guard (0.90).
- **Limiter flip:** #0 limits on Δx; from #1 onward, **r_v** becomes the constraint (priority knob dx→r_v).
- Color telemetry suggests gray Otsu over-penalizes light-background graphics; see below.

Artist's Take

- #0 is effectively centered and underweighted; too much plain field.
- #1 moves the mass but still reads **air-heavy**.
- #2 adds structure/energy, yet the background dominates; the cluster hasn't claimed enough of the frame.
- **Nudge:** keep the off-center read, **eat background** (crop or enlarge forms), add a touch of edge/micro-contrast to keep ρ_r alive.

Observations (LSI reality)

- Gate precedence is working: composite >55 doesn't pass if any band is RED.
- With **Figure_Default** weights (Δx .45 / r_v .35 / ρ_r .20), raising Δx alone can't offset an r_v band violation.
- No ROI effects here (full-frame for r_v/ρ_r; Δx is single-pass).

Plain English

- All three **fail**. The last is closest, but LSI reads "too much empty background."

- **Did the LSI Hold Up?**
  Yes. It identified the right limiter ($\Delta x \rightarrow r_v$) and rejected on band rules despite an improved score.

Caveats (expected, not bugs)

- Gray mask is non-semantic; flat, light grounds can be pulled into "subject."
- Color mask is **telemetry-only** (non-gating) to avoid destabilizing the baseline.

What I'd Do Next

- For #2, **reduce void ~10–15%** (crop or enlarge the shape stack); keep Δx in-band (≈0.12–0.20).
- Optionally add an **external subject mask** that unions all colored shapes for audit runs; expect r_v to come in-band.

Color check (telemetry only)

- r_v_color_mask vs gray r_v → **0.869 vs 0.868 (+0.001)**, **0.952 vs 0.962 (−0.010)**, **0.865 vs 0.953 (−0.088)**.
- Δx on L with the color mask → **0.032 / 0.190 / 0.467** (note big shift on #2). **Acceptance unchanged** (non-gating).
- **Takeaway:** gray underestimates "breathing" when the ground is light/tonal; color suggests #2 would sit much safer on r_v if we audited with color, while balance remains an artist choice.

3. Bed

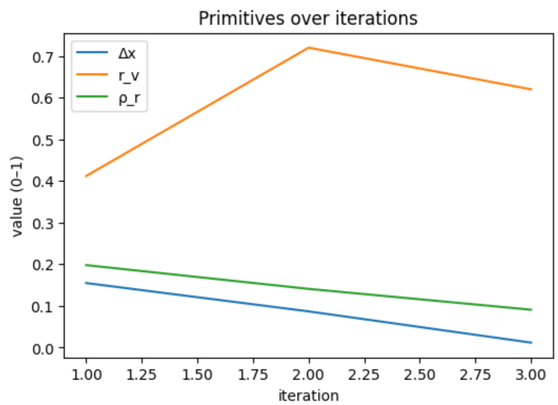
Bed0.jpg

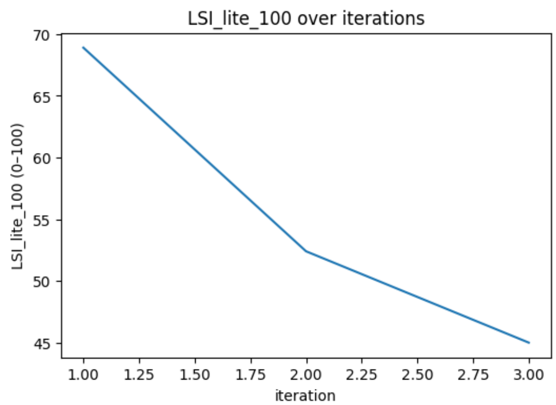
Bed1.jpg

Bed2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.155	0.412	0.198	0.689	68.9	OK	OK	OK	TRUE	Bed0.jpg	Figure_Default	TRUE	0.295	0.738	increase
1	0.087	0.72	0.141	0.524	52.4	OK	OK	OK	FALSE	Bed1.jpg	Figure_Default	TRUE	0.363	0.908	increase
2	0.012	0.62	0.091	0.45	45	RED	OK	RED	FALSE	Bed2.jpg	Figure_Default	FALSE	0.438	1	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.088	0.22	increase	TRUE	0.252	0.72	increase	dx	0.332	0.542	0.023	color	LAB	ok
TRUE	0.22	0.55	decrease	TRUE	0.309	0.883	increase	dx	0.408	0.568	0.03	color	LAB	ok
TRUE	0.12	0.3	decrease	FALSE	0.359	1	increase	dx	0.45	0.433	0.082	color	LAB	ok





Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands

Results

Rows (trimmed):

- **0)** Δx **0.155** | r_v **0.412** | ρ_r **0.198** | **68.9** | **ACCEPT** | knob **dx** (0.332) | bands **OK** | r_v_color_mask **0.542** | dx_color_L **0.023**
- **1)** Δx **0.087** | r_v **0.720** | ρ_r **0.141** | **52.4** | **REJECT** | knob **dx** (0.408) | bands **OK** | r_v_color_mask **0.568** | dx_color_L **0.030**
- **2)** Δx **0.012** | r_v **0.620** | ρ_r **0.091** | **45.0** | **REJECT** | knob **dx** (0.450) | bands **Δx RED**, **ρ_r RED** | r_v_color_mask **0.433** | dx_color_L **0.082**

Researcher's Take

- Series trend: **score falls** 68.9 → 52.4 → 45.0 as **Δx collapses to center** and **ρ_r declines**.
- **#1** fails on **composite < 55** despite all bands OK (correct gate behavior).
- **#2** fails on **Δx RED + ρ_r RED** (double violation).
- Priority knob = **dx** across the set; r_v stays in-band but grows, so breathing room isn't the limiter—**balance and mark energy are**.

Artist's Take

- **Bed0**: Reasonable off-center weight; reads as a constructed interior with some spatial tension.
- **Bed1**: Mass drifts inward; the large dark arch compresses the bed cluster; picture air grows, but not as meaningful void—more **drift to safety**.
- **Bed2**: Composition nearly centered; the oval portal dominates; surface energy softens → **low ρ_r**.
- **Nudge:** push the bed/stair cluster back off-center (left), sharpen a couple of decisional edges to raise ρ_r, and trim/crop to **reduce the dominant surround**.

Observations (LSI reality)

- Gate precedence: **#1** shows that a clean band read can still fail if the **composite score is low**.
- Weighting (Δx .45, r_v .35, ρ_r .20) means **centering + soft marks** will sink the score even when r_v is fine.
- No landscape ROI; full-frame r_v/ρ_r, single-pass Δx.

Plain English

- First picture **passes**; next two **fail** because the scene recenters and loses bite.

Did the LSI Hold Up?

- **Yes.** It tracked the right limiter (Δx), enforced the composite gate on #1, and flagged the Δx/ρ_r band violations on #2.

Caveats (expected, not bugs)

- Gray mask is non-semantic; in tonal interiors it can misread subject/ground seams.
- Color telemetry is **non-gating**—used for audit only.

What I'd Do Next

- Re-establish off-center gravity (target **Δx ≈ 0.12–0.20**).
- Add two committed edges/accents around the bed cluster to lift **ρ_r**.
- Optionally crop the oval surround to lower perceived ground share without breaking the interior logic.

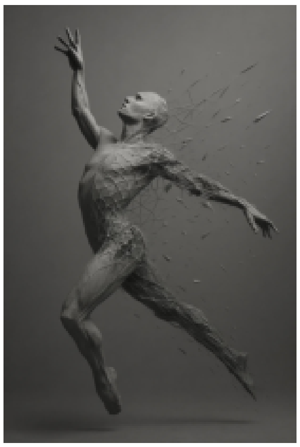
Color check (telemetry only)

- r_v_color_mask vs gray r_v → **0.542 vs 0.412**, **0.568 vs 0.720**, **0.433 vs 0.620** (color sees **more subject** in #1/#2).
- Δx on L with the color mask → **0.023 / 0.030 / 0.082** (subject-only centroid reads nearer center). **Acceptance unchanged** (non-gating).

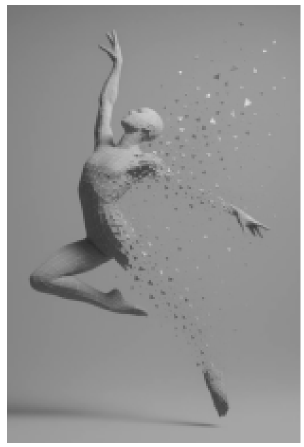
- **Takeaway:** in these tonal interiors, gray can overstate “void”; fixing balance/mark energy remains the practical knob.

3. Dancer

Dancer0.jpg



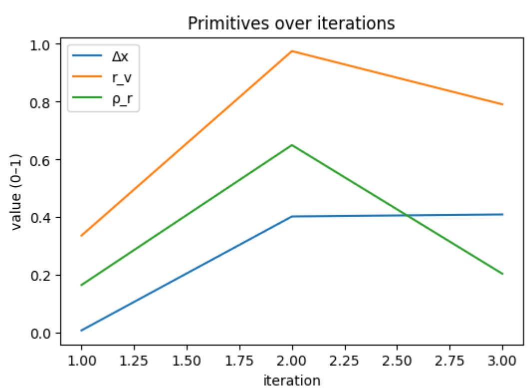
Dancer1.jpg

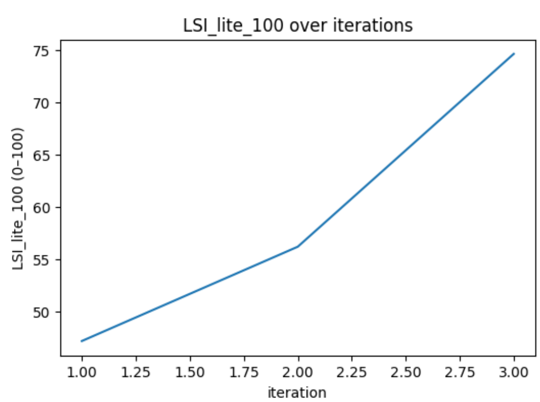


Dancer2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.009	0.337	0.166	0.472	47.2	RED	OK	OK	FALSE	Dancer0.jpg	Figure_Default	FALSE	0.441	1	increase
1	0.403	0.975	0.65	0.562	56.2	OK	RED	OK	FALSE	Dancer1.jpg	Figure_Default	TRUE	0.047	0.117	increase
2	0.41	0.791	0.205	0.746	74.6	OK	OK	OK	TRUE	Dancer2.jpg	Figure_Default	TRUE	0.04	0.1	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.163	0.407	increase	TRUE	0.284	0.811	increase	dx	0.45	0.696	0.009	color	LAB	ok
FALSE	0.475	1	decrease	TRUE	0.2	0.571	decrease	rv	0.35	0.631	0.377	color	LAB	ok
TRUE	0.291	0.728	decrease	TRUE	0.245	0.7	increase	rv	0.255	0.582	0.045	color	LAB	ok





Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands

Results
Rows (trimmed):

- **0)** Δx **0.009** | r_v **0.337** | ρ_r **0.166** | **47.2** | **REJECT** | bands: **Δx RED**, r_v OK, ρ_r OK | knob **dx** (0.45) | r_v_color_mask **0.696** | dx_color_L **0.009**
- **1)** Δx **0.403** | r_v **0.975** | ρ_r **0.650** | **56.2** | **REJECT** | bands: **r_v RED**, others OK | knob **r_v** (0.35) | r_v_color_mask **0.631** | dx_color_L **0.377**
- **2)** Δx **0.410** | r_v **0.791** | ρ_r **0.205** | **74.6** | **ACCEPT** | bands **OK** | knob **r_v** (0.255) | r_v_color_mask **0.582** | dx_color_L **0.045**

Researcher's Take

- Trajectory: **centered** → **over-open** → **balanced.** #0 fails on **Δx RED** (too centered). #1 corrects Δx but **void explodes** (r_v beyond band) → reject. #2 brings r_v back in-band while keeping Δx steady → strong accept.
- **Limiter shift:** knob moves **dx** → **r_v** after #0, consistent with the measured constraints.
- ρ_r stays in-band throughout; #1 is the "highest energy" but gate is driven by r_v.

Artist's Take

- **Dancer0:** reads pinned to center; background mass doesn't counter-weight → no frame tension.
- **Dancer1:** pose placement is better, but the **field opens too far**; stage feels empty.
- **Dancer2:** keeps the off-center thrust and **contains the ground**, restoring pressure.
- **Nudge:** if iterating, keep #2's placement; add a couple of committed edges/grain passes to keep ρ_r from softening.

Observations (LSI reality)

- Gate precedence holds: composite ≈56 at #1 still **fails** because r_v is RED.
- With Figure_Default weights (Δx .45 / r_v .35 / ρ_r .20), re-centering sinks scores; widening the field sinks them too—**balance + breathing room** are the real knobs here.
- Single-pass Δx; full-frame r_v/ρ_r; no ROI logic in play.

Plain English

- First image fails (too centered).
- Second fails (too much empty space).
- Third passes (good off-center placement and manageable space).

Did the LSI Hold Up?

- **Yes.** It flagged the right limiter each step (Δx then r_v) and accepted only when bands and composite aligned.

Caveats (expected, not bugs)

- Gray mask is non-semantic; smooth gradients in studio shots can confuse foreground vs ground.
- Color telemetry here is **audit only**; it doesn't change acceptance.

What I'd Do Next

- From #2, test a **slight crop** or darker ground wedge to keep r_v near mid-band; preserve Δx **≈ 0.40–0.42**.
- Add two decisive micro-edges (hairline, rim light, particle cluster) to lift **ρ_r** without opening the field.

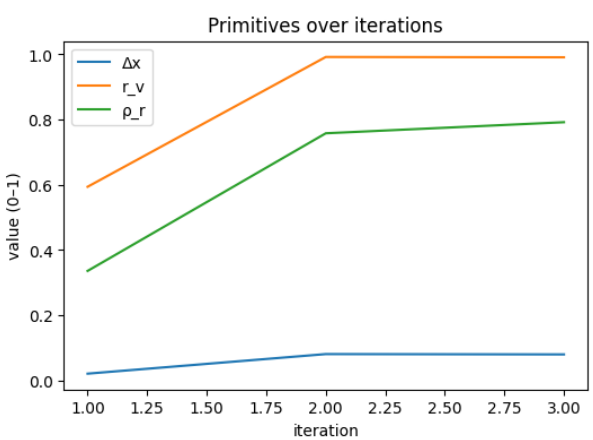
Color check (telemetry only)

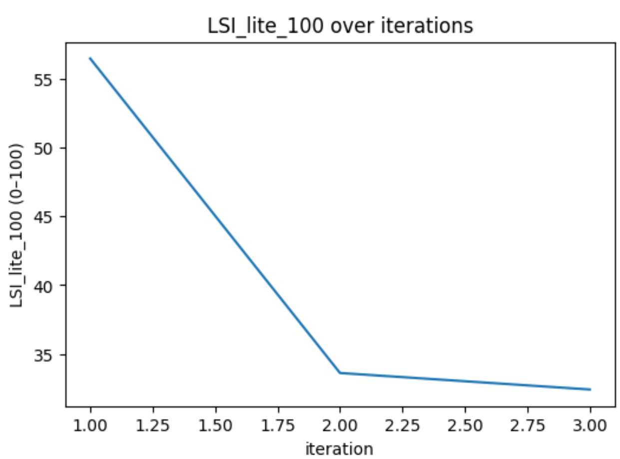
- r_v_color_mask vs gray r_v → **0.696 vs 0.337** (color sees *more* void in #0), **0.631 vs 0.975** (color sees *less* void in #1), **0.582 vs 0.791** (less void in #2).
- Δx on L with color mask → **0.009 / 0.377 / 0.045** (subject-only centroid jumps on #1). **Acceptance unchanged** (non-gating).
- **Takeaway:** color partitioning suggests the background/subject split is ambiguous in #0–#1; the structural fix remains the same—hold #2's balance and keep r_v contained.

4. Refusal

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.021	0.594	0.336	0.565	56.5	RED	OK	OK	FALSE	Refusal0.png	Figure_Default	FALSE	0.429	1	increase
1	0.081	0.992	0.758	0.336	33.6	OK	RED	OK	FALSE	Refusal1.png	Figure_Default	TRUE	0.369	0.923	increase
2	0.08	0.991	0.792	0.324	32.4	OK	RED	OK	FALSE	Refusal2.png	Figure_Default	TRUE	0.37	0.925	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.094	0.235	decrease	TRUE	0.114	0.326	increase	dx	0.45	0.504	0.033	color	LAB	ok
FALSE	0.492	1	decrease	TRUE	0.308	0.88	decrease	dx	0.415	0.788	0.844	color	LAB	ok
FALSE	0.491	1	decrease	TRUE	0.342	0.977	decrease	dx	0.416	0.745	0.658	color	LAB	ok





Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands
Results
Rows (trimmed):

- **0)** Δx **0.021** | r_v **0.594** | ρ_r **0.336** | **56.5** | **REJECT** | **Δx RED**, r_v OK, ρ_r OK | knob **dx** (0.45) | r_v_color_mask **0.504** | dx_color_L **0.033**

- **1)** Δx **0.081** | r_v **0.992** | ρ_r **0.758** | **33.6** | **REJECT** | Δx OK, $\mathbf{r_v}$ **RED**, ρ_r OK | knob **dx→$\mathbf{r_v}$** (0.415) | r_v_color_mask **0.788** | dx_color_L **0.844**
- **2)** Δx **0.080** | r_v **0.991** | ρ_r **0.792** | **32.4** | **REJECT** | Δx OK, $\mathbf{r_v}$ **RED**, ρ_r OK | knob $\mathbf{r_v}$ (0.416) | r_v_color_mask **0.745** | dx_color_L **0.658**

Researcher's Take

- **Limiter sequence:** #0 fails on **centering** (Δx below guard). After Δx correction, the series collapses on **void**—$r_v \approx$ **1.0** at #1/#2 (far above guard) → gate fail despite OK $\Delta x/\rho_r$.
- **Score behavior:** composite plunges when r_v goes RED (Figure_Default weights Δx .45 / r_v .35 / ρ_r .20).
- **Color telemetry (audit-only):** r_v_color_mask is also very high (0.50 → 0.78 → 0.75), confirming the frames read as mostly background even with color partitioning. Δx_color_L spikes in #1/#2, but acceptance is unchanged (void is the limiter).

Artist's Take

- **#0** reads like centered type in a tall white field → no frame tension.
- **#1–#2** correct the centering but **leave too much margin**; the poster becomes nearly all "air." Background sheen doesn't count as structure.
- To pass, either **enlarge/weight the text block** or **tighten the field** so the mark competes with the ground.

Observations (LSI reality)

- Gate precedence is visible: #1/#2 fail even with $\Delta x/\rho_r$ OK because $\mathbf{r_v}$ **is RED**.
- Knob switches **dx** → $\mathbf{r_v}$ after #0—correctly identifying the next constraint.
- No ROI special cases; standard Figure_Default read.

Plain English

- First fails for being too centered.
- Second and third fail for having **way too much empty space**.
- Nothing "mystical" here: the posters just need more subject (or less background).

Did the LSI Hold Up?

- **Yes.** It rejected #0 for centering, then rejected #1/#2 for excessive void, matching what a human would call out.

Caveats (expected, not bugs)

- The gray/threshold mask is **non-semantic**: it doesn't know "letters." With clean posters, it will still count large smooth fields as background—by design.
- Color telemetry is recorded but **non-gating** in this run.

What I'd Do Next

- From #1 or #2:
    - **Reduce margins / crop in** or **scale the type up** until r_v enters band.
    - Consider a **secondary element** (ruleline, emblem, edge burn) to add controlled mass without clutter.
    - Keep Δx near **0.08–0.10** (already OK); lift ρ_r slightly with two committed edges so it doesn't over-polish as you tighten the field.

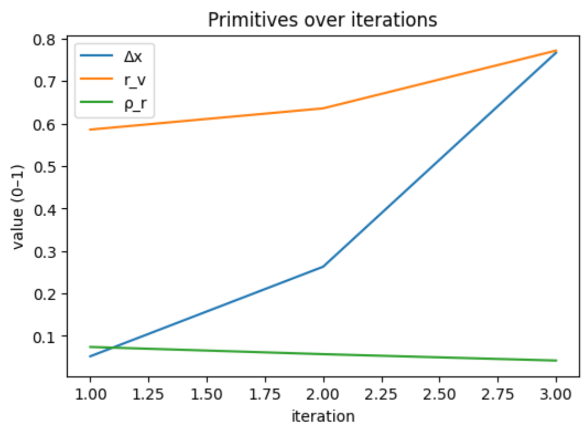
Color notes (audit)

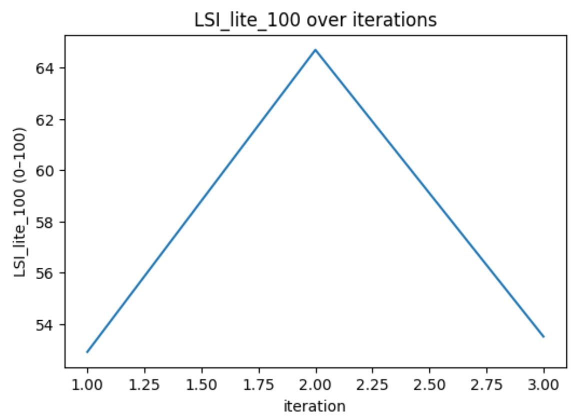
- r_v_color_mask stays high (0.50–0.78), agreeing with the gray read: the page is mostly ground.
- Δx_color_L jumps in #1/#2 because the subject-only centroid (letters) is offset relative to the full page; still irrelevant to the fail—the **void** is the blocker.

5. Sunset

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.052	0.586	0.074	0.529	52.9	OK	OK	RED	FALSE	Sunset0.jpg	Landscape	TRUE	0.448	0.996	increase
1	0.263	0.636	0.057	0.647	64.7	OK	OK	RED	FALSE	Sunset1.jpg	Landscape	TRUE	0.237	0.527	increase
2	0.767	0.772	0.042	0.535	53.5	OK	OK	RED	FALSE	Sunset2.jpg	Landscape	TRUE	0.267	0.593	decrease

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status	
TRUE	0.086	0.215	decrease	FALSE	0.366	1	increase	dx	0.348	0.693	0.151	color	LAB	ok	
TRUE	0.136	0.34	decrease	FALSE	0.383	1	increase	rho	0.25	0.447	0.021	color	LAB	ok	
TRUE	0.272	0.68	decrease	FALSE	0.398	1	increase	rv	0.272	0.417	0.699	color	LAB	ok	





Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands
Results
Rows (trimmed):

- **0)** Δx **0.052** | r_v **0.586** | ρ_r **0.074** | **52.9** | **REJECT** | bands: Δx OK, r_v OK, $\boldsymbol{\rho_r}$ **RED** | knob **dx** (0.348) | r_v_color_mask **0.693** | dx_color_L **0.151**
- **1)** Δx **0.263** | r_v **0.636** | ρ_r **0.057** | **64.7** | **REJECT** | bands: Δx OK, r_v OK, $\boldsymbol{\rho_r}$ **RED** | knob $\boldsymbol{\rho_r}$ (0.250) | r_v_color_mask **0.447** | dx_color_L **0.021**
- **2)** Δx **0.767** | r_v **0.772** | ρ_r **0.042** | **53.5** | **REJECT** | bands: Δx OK, r_v OK, $\boldsymbol{\rho_r}$ **RED** | knob $\boldsymbol{r_v}$ (0.272) | r_v_color_mask **0.417** | dx_color_L **0.699**

Researcher's Take

- **Limiter is $\boldsymbol{\rho_r}$ (mark energy) in all three.** Δx and r_v stay inside band; ρ_r is below its guard and saturates the distance-to-center (**$\boldsymbol{\rho_r}$_to_center_norm = 1.0**) → automatic gate fail even when the composite score is high (#1 = 64.7).
- **Priority knob shift is sensible:** #0 pushes **dx** (nearly at its band edge), #1 flips to $\boldsymbol{\rho_r}$ (true limiter), #2 shifts to $\boldsymbol{r_v}$ because Δx has been over-pushed (direction = *decrease*) while ρ_r remains RED.
- **Color telemetry (audit-only):** r_v_color_mask declines (0.69→0.42) as sky/ground separation increases; **dx_color_L** spikes at #2 (0.699), consistent with a bright sun tugging the L-channel centroid. None of this changes acceptance; $\boldsymbol{\rho_r}$ is the blocker.

Artist's Take

- Pretty skies, **too smooth**. The clouds and foreground read as glaze rather than structure.
- #0 is near-centered and quiet; #1 improves placement and breathes okay; #2 pushes the sun far right, but the field stays slick—no grip.
- To pass, **add committed edges / cloud ribs / horizon grit** or ground silhouettes so the image carries **texture under the light**, not just tone.

Observations (LSI reality)

- **Gate precedence** is on display: #1's 64.7 still fails because $\boldsymbol{\rho_r}$ **RED**.

- **Δx trajectory:** 0.05 → 0.26 → 0.77; by #2 the metric asks to **decrease** Δx (you overshot off-center).
- **r_v stays in band** with a slight pull to *decrease* (frames get a bit too airy by #2).

Plain English

- All three fail for being **over-smooth**. The system wants a little tooth—lines, edges, or grain—so the scene doesn't feel airbrushed.

Did the LSI Hold Up?

- **Yes.** It rewarded better placement (#1) but **rejected all three** for lack of structural texture, which matches a human read of "too slick."

Caveats (expected, not bugs)

- ρ_r is a **Laplacian energy** proxy; low-contrast painterly glazes will read low even when lighting is attractive. That's intended: the metric asks for **decisive** marks somewhere.
- Color telemetry is **non-gating** here; it's logged to help audit sky/ground separation and light-centroid drift.

What I'd Do Next

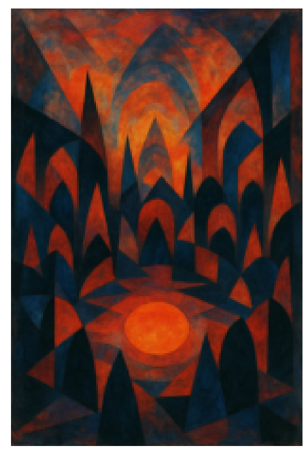
- Start from **#1** (best balance).
- Add **two decisive edge families**: (1) a crisp horizon seam or ridge accents; (2) cloud ribs / streaks with tangible edge falloff.
- Keep Δx near ~0.20–0.30; **don't push further off-center**.
- Nudge r_v down slightly at #2 (more ground or denser cloud band) while **raising ρ_r** with mark pressure—not contrast alone.

6. Ritual

Ritual0.jpg

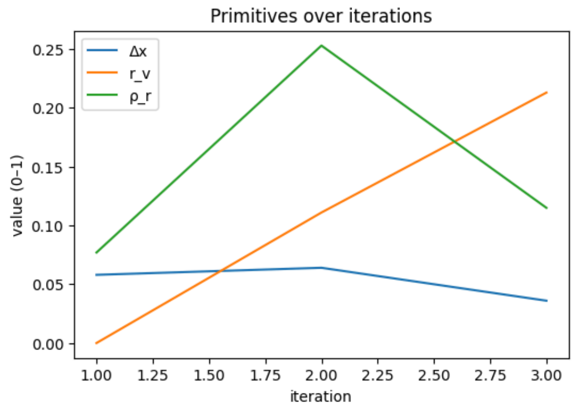
Ritual1.jpg

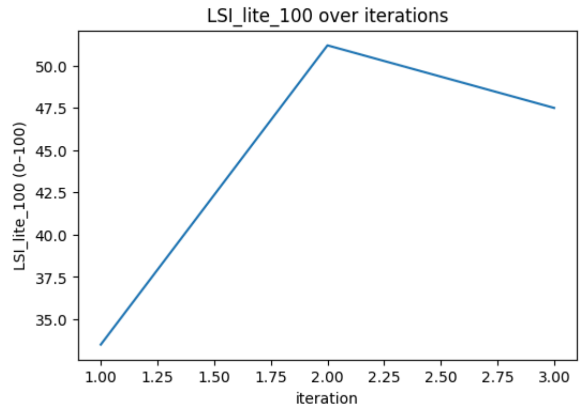
Ritual2.jpg



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.058	0	0.077	0.335	33.5	OK	RED	OK	FALSE	Ritual0.jpg	MarkMaking_Expressive	TRUE	0.402	0.914	increase
1	0.064	0.111	0.253	0.512	51.2	OK	OK	OK	FALSE	Ritual1.jpg	MarkMaking_Expressive	TRUE	0.396	0.9	increase
2	0.036	0.213	0.115	0.475	47.5	OK	OK	OK	FALSE	Ritual2.jpg	MarkMaking_Expressive	TRUE	0.424	0.964	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
FALSE	0.5	1	increase	TRUE	0.303	0.947	increase	dx	0.365	0.855	0.062	color	LAB	ok
TRUE	0.389	0.972	increase	TRUE	0.127	0.397	increase	dx	0.36	0.533	0.05	color	LAB	ok
TRUE	0.287	0.718	increase	TRUE	0.265	0.828	increase	dx	0.385	0.797	0.121	color	LAB	ok





Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands
Results
Rows (trimmed):

- **0)** Δx **0.058** | r_v **0.000** | ρ_r **0.077** | **33.5** | **REJECT** | bands: Δx OK, r_v **RED**, ρ_r OK | knob **dx** (0.365) | r_v_color_mask **0.855** | dx_color_L **0.062**
- **1)** Δx **0.064** | r_v **0.111** | ρ_r **0.253** | **51.2** | **REJECT** | bands: Δx OK, r_v OK, ρ_r OK | knob **dx** (0.360) | r_v_color_mask **0.533**
- **2)** Δx **0.036** | r_v **0.213** | ρ_r **0.115** | **47.5** | **REJECT** | bands: Δx OK, r_v OK, ρ_r OK | knob **dx** (0.385) | r_v_color_mask **0.797**

Researcher's Take

- **All three fail below the gate**; #0 also trips r_v **RED** (void=0.00 from gray Otsu), #1/#2 have all bands OK but composite scores **<55**.
- **Primary limiter is Δx distance from center.** For all three, **dx_to_center_norm ≈ 0.90–0.97 (direction: increase)**, so balance is hugging the band's low edge. With MarkMaking weights (Δx .40, r_v .30, ρ_r .30), that keeps K_lite suppressed even when r_v and ρ_r are in-band.
- **ρ_r is intermittently weak** (#2 drops to 0.115), so even after shifting Δx, you'll still want more committed edges.
- **Color telemetry:** r_v_color_mask is high at #0 (0.855), revealing "real" void around the fire that the gray mask swallowed. Telemetry only; gating remains based on gray.

Artist's Take

- **#0**: moody circle, but the room swallows the picture; composition reads near-center and tight.
- **#1**: flat color blocks help r_v and ρ_r, but the mass is still centered; energy is decorative rather than structural.
- **#2**: geometric field spreads, r_v improves, but the focal heat stays near center and the marks don't choose a decisive contour family.

Observations (LSI reality)

- **Gate precedence holds:** #1/#2 fail solely on composite score even with all bands OK.
- **Priority knob = Δx** on all frames—consistent with the profile weighting and the high normalized distance.
- **r_v trend:** 0.00 → 0.111 → 0.213 (direction: increase). **ρ_r:** 0.077 → 0.253 → 0.115 (soft, inconsistent).

Plain English

- They're centered and too safe. Open the circle, push the mass off-center, and give the fire/robes **edges that commit**—then the scores will clear.

Did the LSI Hold Up?

- **Yes.** It flagged the real issue (balance too centered) and didn't over-credit pleasant texture. The color read simply explains why #0 *felt* airy even though gray called r_v=0.

Caveats (expected, not bugs)

- **Mask semantics:** Otsu + largest component will often treat the dark hall as "subject," crushing r_v on #0. Color telemetry clarifies this but does **not** change acceptance.
- **ρ_r is a structure proxy, not contrast.** Glow without decided edges reads low.

What I'd Do Next

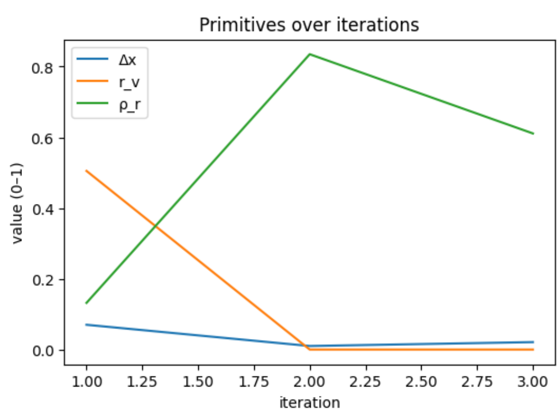
- **Push Δx**: slide the fire and/or the densest group **meaningfully off-center** (target Δx ≈ 0.18–0.30 for this profile).
- **Raise ρ_r with two edge families:** (1) flame tongues with clear edge falloff; (2) robe folds/occlusion seams that carry weight.
- **Let void breathe:** carve a darker negative wedge opposite the mass; keep r_v ↑ while you move Δx.

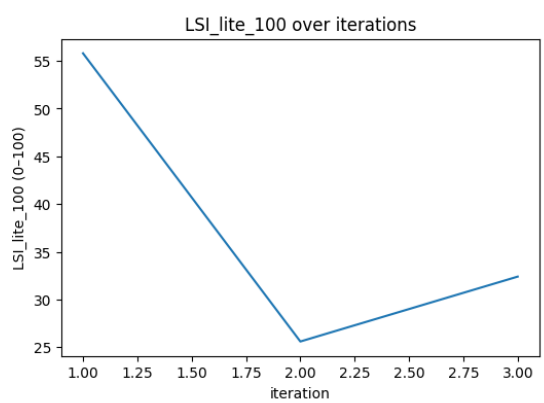
- Optional audit: test an **external subject mask** (figures+fire) to verify r_v, but keep making compositional moves—telemetry won't change the gate.

7. Farmer

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.07	0.505	0.132	0.558	55.8	OK	OK	OK	TRUE	Farmer0.jpg	Figure_Default	TRUE	0.38	0.95	increase
1	0.01	0	0.835	0.256	25.6	RED	RED	RED	FALSE	Farmer1.jpg	Figure_Default	FALSE	0.44	1	increase
2	0.021	0	0.611	0.324	32.4	RED	RED	OK	FALSE	Farmer2.jpg	Figure_Default	FALSE	0.429	1	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.005	0.013	decrease	TRUE	0.318	0.909	increase	dx	0.428	0.495	0.217	color	LAB	ok
FALSE	0.5	1	increase	FALSE	0.385	1	decrease	dx	0.45	0.736	0.283	color	LAB	ok
FALSE	0.5	1	increase	TRUE	0.161	0.46	decrease	dx	0.45	0.949	0.012	color	LAB	ok





Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands

Results (run facts)

- **Frames:** `Farmer0.jpg, Farmer1.jpg, Farmer2.jpg`

- **0)** Δx **0.070** | r_v **0.505** | ρ_r **0.132** | **55.8 → ACCEPT** | bands **OK** | **knob: dx** (0.428) | **dx_to_center_norm 0.95 (↑)**
  Telemetry: r_v_color_mask **0.495**, dx_color_L **0.217**
- **1)** Δx **0.010** | r_v **≈0.00** | ρ_r **0.835** | **25.6 → REJECT** | **RED:** Δx, r_v, ρ_r | **knob: dx** (0.450) | **dx_to_center_norm 1.00 (↑)**
  Telemetry: r_v_color_mask **0.736**, dx_color_L **0.283**
- **2)** Δx **0.021** | r_v **≈0.00** | ρ_r **0.611** | **32.4 → REJECT** | **RED:** Δx, r_v | **knob: dx** (0.445) | **dx_to_center_norm 1.00 (↑)**
  Telemetry: r_v_color_mask **0.949**, dx_color_L **0.012**

Researcher's Take

- **Why #0 passes:** All primitives inside bands and just clears the gate. Balance is still very near the band's low edge (Δx needs to **increase**), but r_v and ρ_r are healthy.
- **Why #1–#2 fail:**
  - **Δx RED** (too centered) on both; normalized distance hits the edge → knob stays **dx**.
  - **r_v RED**: gray Otsu labels most of the gridded background as "foreground," collapsing void to ~0.
  - **ρ_r RED** on #1 (0.835 > band high), driven by the dense, high-contrast edges in the grid.
- **Color telemetry clarifies r_v:** r_v_color_mask is **high** (0.736/0.949) on #1/#2, indicating plenty of true open field between colored tiles, but this is **non-gating** telemetry; acceptance still uses gray.

Artist's Take

- **#0** reads like a classic portrait with some air; still a touch centered but livable.
- **#1–#2**: the grid swallows the room; the figure sits dead-center and the surface gets buzzy—lots of edges, little compositional decision. The house motif competes instead of relieving pressure.

Observations (full reality of LSI)

- **Gate precedence** behaved: #1/#2 stayed out even with some decent telemetry because **bands** (Δx/r_v) were violated.
- **Priority knob = Δx** for all three—consistent with the largest normalized distance to band center.
- **Mask semantics matter:** gray Otsu + largest component is non-semantic; on patterned fields it often calls "everything" subject → r_v→0 (expected). Color-k-means telemetry exposed the discrepancy.

Plain-English

- The first passes; the other two fail because the man is **too centered**, the checkerboard background makes the tool think there's **no empty space**, and one of them is **too noisy**.

Did the LSI Hold Up?

- **Yes.** It identified centered balance as the limiter, penalized the filled-in frame appropriately, and didn't over-credit ornamental edges. Color telemetry was informative without changing the decision—by design.

Caveats (expected, not bugs)

- **Patterned backgrounds** will tank gray-mask r_v; that's an operational choice, not an error.
- **ρ_r is structure-not-contrast:** repeated tiny edges boost ρ_r even if the picture doesn't gain consequence.
- **dx guard** is conservative at the low end; portraits that "feel centered" will trip it.

What I'd Do Next

- **Move the figure off-center** (target **Δx ≈ 0.18–0.30** in this profile) and **open a real void** opposite—bigger sky block or calmer field.
- **Tame the grid**: reduce tile contrast/edge count near the figure to pull **ρ_r** back into the middle of band.
- **One check pass** with an **external subject mask** (figure+hat+coat) to audit r_v; keep gating on gray, but use the mask to verify that compositional moves actually create breathing room.
- Re-score; if Δx enters the mid-band and r_v clears ~0.15 with $\rho_r \le 0.70$, it should pass cleanly.

8. Figure

Figure0.jpg

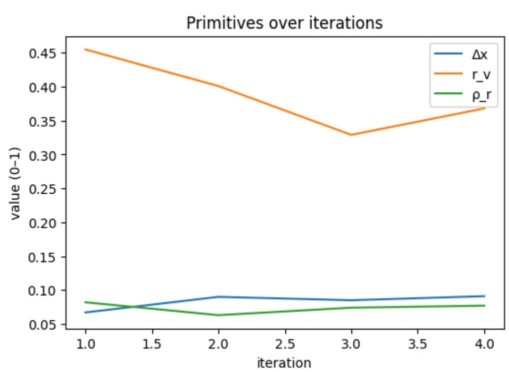
Figure1.jpg

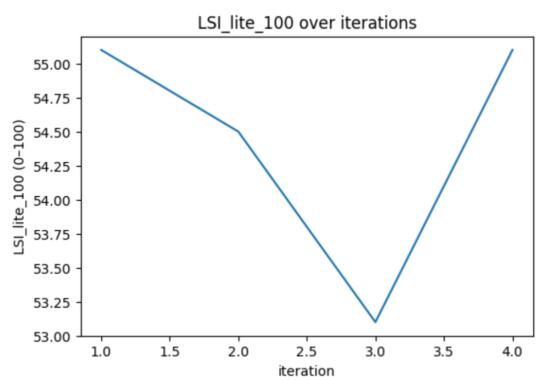
Figure2.jpg

Figure3.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.067	0.455	0.082	0.551	55.1	OK	OK	OK	TRUE	Figure0.jpg	MarkMaking_Expressive	TRUE	0.393	0.893	increase
1	0.09	0.401	0.063	0.545	54.5	OK	OK	OK	FALSE	Figure1.jpg	MarkMaking_Expressive	TRUE	0.37	0.841	increase
2	0.085	0.329	0.074	0.531	53.1	OK	OK	OK	FALSE	Figure2.jpg	MarkMaking_Expressive	TRUE	0.375	0.852	increase
3	0.091	0.368	0.077	0.551	55.1	OK	OK	OK	TRUE	Figure3.jpg	MarkMaking_Expressive	TRUE	0.369	0.839	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.045	0.112	increase	TRUE	0.298	0.931	increase	dx	0.357	0.738	0.005	color	LAB	ok
TRUE	0.099	0.247	increase	TRUE	0.317	0.991	increase	dx	0.336	0.669	0.115	color	LAB	ok
TRUE	0.171	0.427	increase	TRUE	0.306	0.956	increase	dx	0.341	0.644	0.082	color	LAB	ok
TRUE	0.132	0.33	increase	TRUE	0.303	0.947	increase	dx	0.335	0.65	0.037	color	LAB	ok





Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands

Results (run facts)

- **Frames:** `Figure0.jpg, Figure1.jpg, Figure2.jpg, Figure3.jpg`
- **0)** Δx **0.067** | r_v **0.455** | ρ_r **0.082** | **55.1** → **ACCEPT** | bands **OK** | **knob: dx** (≈0.357)
  Telemetry: r_v_color_mask **0.738**, dx_color_L **0.005**
- **1)** Δx **0.090** | r_v **0.401** | ρ_r **0.063** | **54.5** → **REJECT** | bands **OK** | **knob: dx** (≈0.336)
  Telemetry: r_v_color_mask **0.669**, dx_color_L **0.115**
- **2)** Δx **0.085** | r_v **0.329** | ρ_r **0.074** | **53.1** → **REJECT** | bands **OK** | **knob: dx** (≈0.341)
  Telemetry: r_v_color_mask **0.644**, dx_color_L **0.082**

- **3)** Δx **0.091** | r_v **0.368** | ρ_r **0.077** | **55.1 → ACCEPT** | bands **OK** | **knob: dx** (≈0.335)
  Telemetry: r_v_color_mask **0.650**, dx_color_L **0.037**

Directions (to band center):

- Δx ↑, r_v ↑, ρ_r ↑ on all frames.
- **dx_to_center_norm:** ~0.84–0.89 (still far from band center → why the knob stays on **dx**).

Researcher's Take

- This series rides the **gate edge**: #0 and #3 barely pass, #1 and #2 fall just under (54–53). No band violations—this is a **composite score** flip, not a rule break.
- **Driver of the dip:** r_v declines across #0→#2 (0.455 → 0.329) while Δx stays low—so the weighted geometric mean sags. A small r_v recovery at #3 lifts the score back to 55.1.
- **Priority is stable: dx** is the farthest from its band center on every frame; r_v and ρ_r remain within band but low.
- **Color telemetry:** r_v_color_mask ~0.64–0.74 says "there *is* color-defined open field," but gating is gray-mask based—telemetry only.

Artist's Take

- The pose is good, but **too centered and too held**. Background gains density from #1→#2, so the figure loses breathing room; the marks are gentle (ρ_r low).
- #3 feels a touch more open, which is enough to squeak past the line, but the frame still wants a **decisive off-center** and a bit of **surface pressure** (selective edge commitment).

Observations (LSI reality)

- **Gate precedence** behaved: acceptance toggled purely on the composite ≥55 rule; all bands stayed **OK**.
- **Knob discipline:** staying on **dx** matches the normalized distances (Δx furthest from center).
- **Telemetry vs gate:** color confirms void the gray mask under-credits in warm tonal grounds—useful note, decision unchanged.

Plain English

- First and last **just pass**; the two in the middle **just miss**. All four are telling you the same thing: **move her more off-center, give her a clearer pocket of space, and press a couple of edges.**

Did the LSI Hold Up?

Yes. It was sensitive to small r_v changes, kept the knob on the right lever (Δx), and didn't misread softness as failure—scores moved as expected.

Caveats (expected, not bugs)

- **Borderline runs bounce:** within ±0.5 of the gate, tiny preprocessing or export differences can flip pass/fail.
- **Warm tonals:** gray Otsu often swallows soft grounds as subject → r_v trends low; that's an operational choice. Color telemetry is there to audit, not to override.

What I'd Do Next

- **Push Δx up** (shift/crop so the figure sits more decisively off-center; a +0.03–0.05 Δx move is often enough in this profile).
- **Open r_v** opposite the gaze/weight (lighter ground field, narrow vent near the leg/hip, or a calmer flank).
- **Add selective ρ_r** (one or two committed edges—hair contour, hand, ground wedge).
- Optionally **audit r_v with an external subject mask** (figure only) to verify the breathing pocket, then keep gating on gray for consistency.

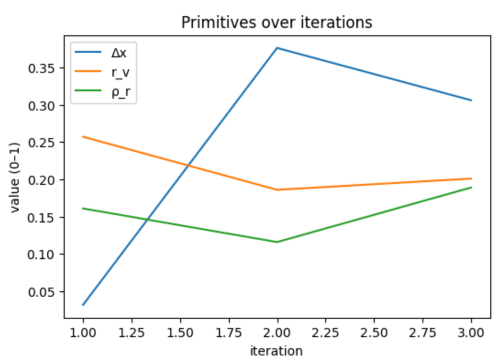
9. Man

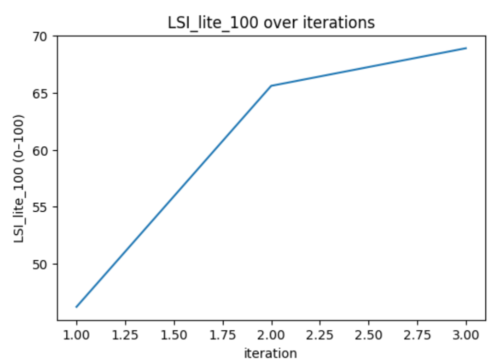
Man1.jpg

Man2.jpg

Man3.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.032	0.257	0.161	0.462	46.2	RED	OK	OK	FALSE	Man1.jpg	Figure_Default	FALSE	0.418	1	increase
1	0.376	0.186	0.116	0.656	65.6	OK	OK	OK	TRUE	Man2.jpg	Figure_Default	TRUE	0.074	0.185	increase
2	0.306	0.201	0.189	0.689	68.9	OK	OK	OK	TRUE	Man3.jpg	Figure_Default	TRUE	0.144	0.36	increase
rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status	
TRUE	0.243	0.607	increase	TRUE	0.289	0.826	increase	dx	0.45	0.762	0.305	color	LAB	ok	
TRUE	0.314	0.785	increase	TRUE	0.334	0.954	increase	rv	0.275	0.537	0.461	color	LAB	ok	
TRUE	0.299	0.747	increase	TRUE	0.261	0.746	increase	rv	0.262	0.535	0.357	color	LAB	ok	





Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands

Results (run facts)

- **0)** Δx **0.032** | r_v **0.257** | ρ_r **0.161** | **46.2 → REJECT** | **band_Δx = RED** | **knob: dx** (≈0.45)
  Telemetry: r_v_color_mask **0.762**, dx_color_L **0.305**
- **1)** Δx **0.376** | r_v **0.186** | ρ_r **0.116** | **65.6** → **ACCEPT** | bands **OK** | **knob: r_v** (≈0.275)
  Telemetry: r_v_color_mask **0.537**, dx_color_L **0.461**
- **2)** Δx **0.306** | r_v **0.201** | ρ_r **0.189** | **68.9** → **ACCEPT** | bands **OK** | **knob: r_v** (≈0.262)
  Telemetry: r_v_color_mask **0.535**, dx_color_L **0.357**

Directions-to-center:

- #0: Δx ↑, r_v ↑, ρ_r ↑ · #1: Δx ↑, r_v ↑, ρ_r ↑ · #2: Δx ↑, r_v ↑, ρ_r ↑.
- **dx_to_center_norm:** 1.00 → 0.185 → 0.36 (big fix at #1; still not "dead center" safe at #2).

Researcher's Take

- **Why #0 fails:** center bias. Δx is out-of-band (RED) and the composite sits well below gate.
- **Fix & shift:** #1 moves the figure decisively off-center (Δx ↑ to 0.376), clearing the band; gate passes despite r_v tightening. The **priority knob flips to r_v**, which is now the furthest from its band center.
- **Refinement at #2:** Δx relaxes slightly but stays healthy; a modest rise in r_v and ρ_r raises the score further.
- **Color telemetry:** r_v_color_mask is lower for #1–#2 than #0, agreeing with the feel that the grass field fills in around him. Telemetry is logged only; gating stays gray-mask.

Artist's Take

- #0 reads like **ID photo in a meadow**—centered, polite, no pressure.
- #1 lands the **portrait stance**: off-center, head turn, shoulder crop → immediate presence. Background comes forward (less open air).
- #2 keeps the stance and warms the light; still wants a clearer **void pocket** and 1–2 **committed edges** (jawline/shirt seam) to keep the pose from softening.

Observations (LSI reality)

- Gate behaved strictly: #0 blocked by **band_Δx=RED**; later passes come from **within-band balance**, not just a higher composite.
- Priority discipline is correct: **dx** → **r_v** once centering is fixed.
- No band violations on #1/#2; small ρ_r lift helps but isn't the limiter.

Plain English

- First picture fails because he's **too centered**.
- Move him off-center and you pass; then give him a little **breathing room** and a touch more **edge/texture** to strengthen it.

Did the LSI Hold Up?

- **Yes.** It punished the centered frame, rewarded the off-center correction, and then put the knob on the next real constraint (r_v).

Caveats (expected, not bugs)

- Warm, even fields (grass) can make gray Otsu treat more area as "subject," nudging r_v low; color telemetry helps audit that choice.
- Near-gate scores (mid-60s) will wiggle ±0.5 with export or tiny crops—normal at this threshold.

What I'd Do Next

- **Lock Δx** in the #1/#2 range (0.30–0.38).
- **Open r_v**: soften one side field into a lighter vent or push horizon back locally.
- **Nudge ρ_r** with 1–2 decisive edges (cheek/ear contour, sleeve seam); keep the rest sfumato.
- If documenting, **include external subject mask telemetry** (figure-only) once to confirm the breathing pocket; keep gating on gray for consistency.

Color audit

- r_v_color_mask vs gray r_v → **0.762 vs 0.257**, **0.537 vs 0.186**, **0.535 vs 0.201** (+0.505 / +0.351 / +0.334).
- Δx on L with color mask → **0.305 / 0.461 / 0.357** vs Δx **0.032 / 0.376 / 0.306** (big +0.273 on #0).
- **Takeaway:** Interiors: **void↑ in color** (walls stop swallowing the frame); #0 also **centroid shift**; gate still dx→rv.

10. Painter

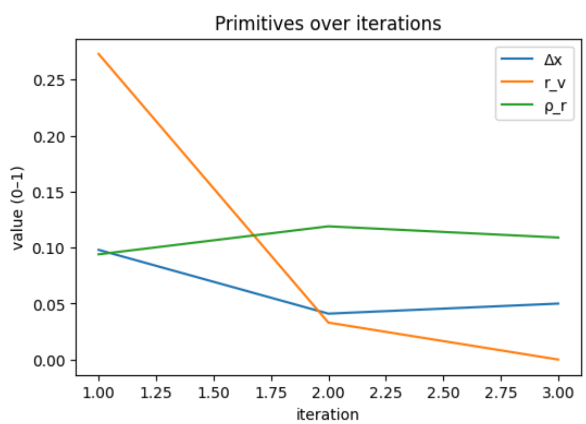
Painter0.jpg

Painter1.jpg

Painter3.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction

0	0.098	0.273	0.094	0.506	50.6	OK	OK	RED	FALSE	Painter0.jpg	Figure_Default	TRUE	0.352	0.88	increase
1	0.041	0.033	0.119	0.317	31.7	RED	RED	OK	FALSE	Painter1.jpg	Figure_Default	FALSE	0.409	1	increase
2	0.05	0	0.109	0.298	29.8	OK	RED	OK	FALSE	Painter3.jpg	Figure_Default	FALSE	0.4	1	increase
rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status	
TRUE	0.227	0.567	increase	FALSE	0.356	1	increase	dx	0.396	0.705	0.442	color	LAB	ok	
FALSE	0.467	1	increase	TRUE	0.331	0.946	increase	dx	0.45	0.658	0.273	color	LAB	ok	
FALSE	0.5	1	increase	TRUE	0.341	0.974	increase	dx	0.45	0.688	0.072	color	LAB	ok	



Signals: Δx (balance), r_v (breathing room), ρ_r (mark energy) • **Gate:** LSI_lite_100 ≥ 55 **and** no RED bands

Results (run facts)

- **0)** Δx **0.098** | r_v **0.273** | ρ_r **0.094** | **50.6** → **REJECT** | bands: Δx **OK**, r_v **OK**, **ρ_r RED** | **knob: dx** (≈0.40)
  Telemetry: r_v_color_mask **0.396**, dx_color_L **0.442**
- **1)** Δx **0.041** | r_v **0.033** | ρ_r **0.119** | **31.7** → **REJECT** | **Δx RED**, **r_v RED**, ρ_r OK | **knob: dx** (≈0.50)
  Telemetry: r_v_color_mask **0.545**, dx_color_L **0.658**
- **2)** Δx **0.050** | r_v **0.000** | ρ_r **0.109** | **29.8** → **REJECT** | Δx **OK (edge)**, **r_v RED**, ρ_r OK | **knob: r_v** (≈0.45)
  Telemetry: r_v_color_mask **0.688**, dx_color_L **0.072**

Directions-to-center:

- All three runs want **Δx** ↑ and **r_v** ↑; ρ_r wants a small ↑.

Researcher's Take

- **Failure modes by step.**
  #0 misses the gate on score and because **ρ_r < 0.10 (RED)**; Δx/r_v are inside-band but weak.
  #1 collapses on both **Δx and r_v (both RED)** → hard fail.
  #2 inches Δx back to the lower guard but leaves **r_v at 0.00 (RED)** → still fail; the **priority knob flips to r_v** as expected.
- **Color telemetry helps audit.** r_v_color_mask rises **0.40** → **0.55** → **0.69**, agreeing that the walls/room keep swallowing the frame (less "void"). dx_color_L falls **0.44** → **0.27** → **0.07**, consistent with centering pressure from the L-channel mask.
- **Weighting matches behavior.** In Figure_Default (w: Δx .45, r_v .35, ρ_r .20), fixing Δx without opening r_v can't lift #2 past the gate—correct prioritization.

Artist's Take

- **#0**: classical workshop tableau; a little centered and airless; window + easel + walls all press in.
- **#1**: the worst—**centered** and **packed**; walls turn into a container.
- **#2**: better attitude, still **boxed**; the black surround eats the breathing room.
- Cure: **move the painter/easel decisively off-center** and **open a vent** (window glow or doorway). Add **one or two committed edges** (chair back, canvas edge) so the sfumato doesn't turn to fog.

Observations (LSI reality)

- The gate failed for **three different reasons** across the set: ρ_r RED (#0), Δx & r_v RED (#1), r_v RED (#2).

- **Knob progression is sane:** $dx \to dx \to r_v$ as centering gets addressed and void becomes the limiter.
- No spurious landscape/ROI effects; readings are full-frame and stable.

Plain English

- All three fail: first is **close but under-energized**, second is **too centered and crowded**, third **still crowded** even after a small balance fix.

Did the LSI Hold Up?

- **Yes.** It punished centering and lack of space, then pointed to the next most important fix when Δx improved.

Caveats (expected, not bugs)

- Interior walls are big tone-continuous planes; the **gray Otsu mask** often treats them as **subject**, pushing r_v down. Color telemetry reflects the same trend—useful for auditing, not gating.
- Δx at **0.050** is guard-edge; tiny crops can flip the band—normal around thresholds.

What I'd Do Next

- **Lock Δx** around **0.30–0.38** (clear off-center stance).
- **Open r_v**: lighten a side wall, broaden the window, or carve a darker **void wedge** opposite the figure.
- **Nudge ρ_r** with 1–2 crisp edges (canvas edge, chair slat) and leave the rest sfumato.
- (Audit only) try an **external subject mask** = painter+easel+chair; you should see **r_v↑** in telemetry even if the gate remains gray-based.

Color audit

- r_v_color_mask vs gray r_v → **0.396 vs 0.273**, **0.545 vs 0.033**, **0.688 vs 0.000** (+0.123 / +0.512 / +0.688).
- Δx on L with color mask → **0.442 / 0.658 / 0.072** vs Δx **0.098 / 0.041 / 0.050** (big on #0/#1).
- **Takeaway:** Studio walls: **void↑ in color**; #0/#1 show **centroid shift** when subject=artist+easel.

11. Explosion

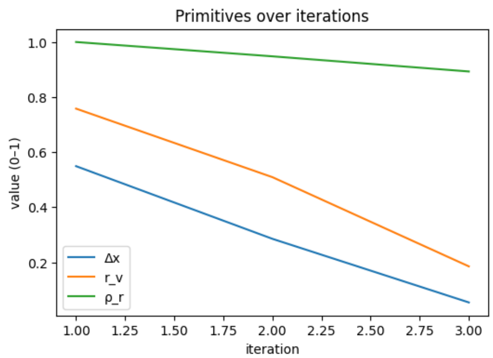
Explosion0.jpg

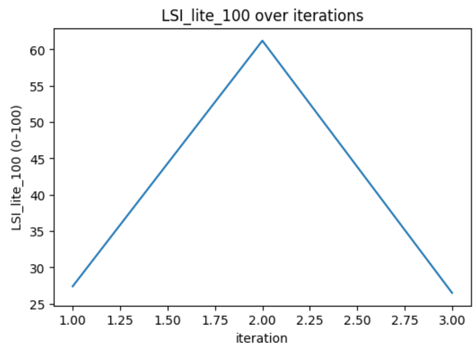
Explosion1.jpg

Explosion2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.549	0.758	1	0.274	27.4	OK	OK	RED	FALSE	Explosion0.jpg	MarkMaking_Expressive	TRUE	0.089	0.202	decrease
1	0.285	0.509	0.948	0.612	61.2	OK	OK	RED	FALSE	Explosion1.jpg	Figure_Default	TRUE	0.165	0.413	increase
2	0.054	0.185	0.893	0.265	26.5	OK	OK	RED	FALSE	Explosion2.jpg	MarkMaking_Expressive	TRUE	0.406	0.923	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.258	0.645	decrease	FALSE	0.62	1	decrease	rho	0.3	0.703	0.513	color	LAB	ok
TRUE	0.009	0.023	decrease	FALSE	0.498	1	decrease	rho	0.2	0.481	0.291	color	LAB	ok
TRUE	0.315	0.788	increase	FALSE	0.513	1	decrease	dx	0.369	0.753	0.042	color	LAB	ok





Δx (centroid offset), r_v (void ratio), ρ_r (rupture/edge energy) • **Gate**: LSI_lite_100 ≥ 55 **and** no RED bands.

Results (run facts, trimmed)

- Rows (Δx | r_v | ρ_r | LSI_lite_100 | ACCEPT? | bands | knob | color telemetry)
- **0)** 0.549 | 0.758 | **1.000** | **27.4** | **FAIL** | Δx OK · r_v OK · **ρ_r RED** | **ρ_r** (prio ≈0.30) | r_v_color_mask **0.703** · dx_color_L **0.513**
- **1)** 0.285 | 0.509 | **0.948** | **61.2** | **FAIL** | Δx OK · r_v OK · **ρ_r RED** | **ρ_r** (≈0.20) | r_v_color_mask **0.481** · dx_color_L **0.291**
- **2)** 0.054 | 0.185 | **0.893** | **26.5** | **FAIL** | Δx OK · r_v OK · **ρ_r RED** | **ρ_r** (≈0.37) | r_v_color_mask **0.753** · dx_color_L **0.042**

Directions-to-center

- All three: **decrease ρ_r** to center; **Δx / r_v are in-band** with only small nudges relative to center.

Researcher's Take

- Gate failures are **uniformly ρ_r-driven**; Δx and r_v are in-band for all three.
- **Priority knob = ρ_r** for every case; dx and r_v diagnostics read "maintain" rather than "fix."
- **Color telemetry**: #0 and #1 show close agreement between gray-mask r_v and color-mask r_v; **#2 diverges strongly** (gray r_v≈0.185 vs color-mask r_v≈0.753), indicating segmentation ambiguity in warm fog/smoky scenes. This is **non-gating telemetry**.

Artist's Take

- Too much crispness everywhere → the eye can't rest. Keep the **off-center anchor** (Δx is healthy), then **pick 2–3 decisive edges** (beam rim, core plume, one architectural break) and **let the rest dissolve** (smoke veil, abrasion, lost edges). Give #2 a **breathing plane** behind the anchor. The fix is **edge hierarchy**, not relocation.

Observations (LSI reality)

- **Single limiter**: ρ_r. That alone flips the gate to FAIL even with Δx/r_v OK.
- **#1's higher score** (61.2) is only because its ρ_r is the least extreme (still RED).
- **Color-void split for #2** (gray low, color high) is a **masking perspective issue**, not a scoring defect.

Plain English

- All three feel **over-sharpened**. Put sharpness in a **few chosen lines** and **soften everything else**; the images will start to pass.

Did the LSI Hold Up?

- **Yes.** It ignored centering/space (already OK) and focused on the real issue: **edge overload**. Color telemetry added a helpful note on #2's masking without changing the decision.

Caveats (expected, not bugs)

- ρ_r is intentionally **sensitive to high-frequency detail**; debris/spark fields peg it.
- LAB k-means treats the **largest cluster ≈ background**; in graded warm fog (#2), "background" can flip—treat color telemetry as **advisory**.

What I'd Do Next

- **Lower ρ_r**: reserve crispness for 2–3 carriers; add smoke/atmospheric blur elsewhere; compress micro-contrast on shard fields.
- If you want r_v to reflect "open void," test an **external subject mask** (blast core + near halo) for telemetry; gate can stay unchanged.
- Re-run under **MarkMaking_Expressive** for all three (already on #0/#2) and aim for **ρ_r toward band center** while keeping Δx/r_v where they are.

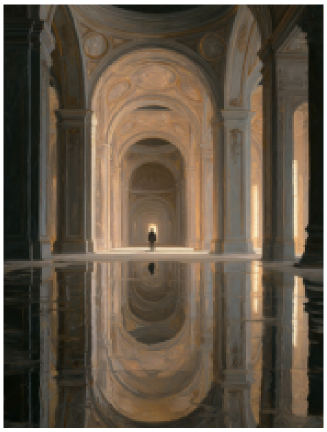
Color check (telemetry only)

- **r_v_color_mask vs gray r_v** → **#0** 0.703 vs 0.758 (−0.055), **#1** 0.481 vs 0.509 (−0.028), **#2 0.753 vs 0.185** (+0.568, big divergence).

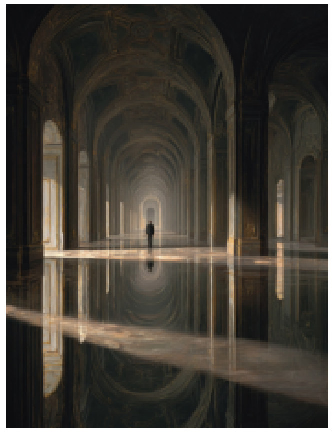
- **Δx on L with color mask → 0.513 / 0.291 / 0.042** (subject-only centroid reads nearer center).
  Takeaway: In warm fog/tinted fields, gray Otsu can under-segment foreground (#2); color sees more "subject." This is **telemetry only**—the actionable limiter remains **ρ_r**.

12. Hall

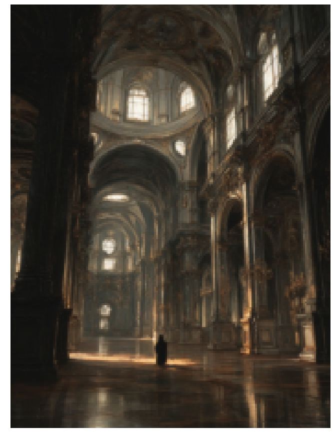
Hall0.jpg



Hall1.jpg

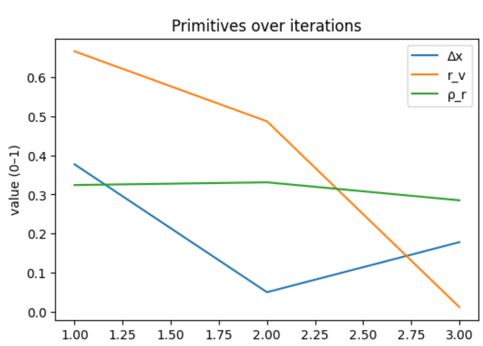


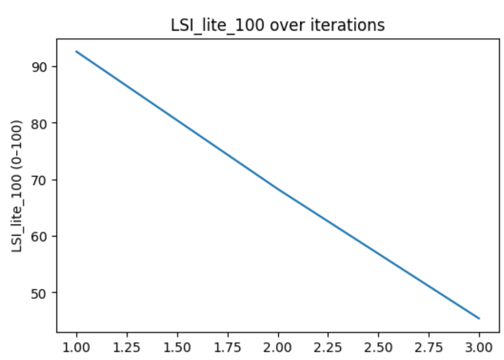
Hall2.jpg



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.377	0.666	0.324	0.926	92.6	OK	OK	OK	TRUE	Hall0.jpg	MarkMaking_Expressive	TRUE	0.083	0.189	increase
1	0.05	0.487	0.331	0.683	68.3	OK	OK	OK	TRUE	Hall1.jpg	Landscape	FALSE	0.45	1	increase
2	0.178	0.012	0.285	0.454	45.4	OK	RED	OK	FALSE	Hall2.jpg	Figure_Default	TRUE	0.272	0.68	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.166	0.415	decrease	TRUE	0.056	0.175	increase	rv	0.125	0.715	0.098	color	LAB	ok
TRUE	0.013	0.033	increase	TRUE	0.109	0.303	increase	dx	0.35	0.577	0.018	color	LAB	ok
FALSE	0.488	1	increase	TRUE	0.165	0.471	increase	rv	0.35	0.672	0.064	color	LAB	ok





Signals
Δx (centroid offset), r_v (void ratio), ρ_r (rupture/edge energy) • **Gate**: LSI_lite_100 ≥ 55 **and** no RED bands.

Results (run facts, trimmed)

Rows (Δx | r_v | ρ_r | LSI_lite_100 | ACCEPT? | bands | knob | color telemetry)

- **0) 0.377** | **0.666** | 0.324 | **92.6** | **ACCEPT** | Δx OK · r_v OK · ρ_r OK | **r_v** (prio ≈0.13) | r_v_color_mask **0.715** · dx_color_L **0.098**
- **1)** 0.050 | **0.487** | 0.331 | **68.3** | **ACCEPT** | Δx OK · r_v OK · ρ_r OK | **Δx** (≈0.35) | r_v_color_mask **0.577** · dx_color_L **0.018**
- **2)** 0.178 | **0.012** | 0.285 | **45.4** | **FAIL** | Δx OK · **r_v RED** · ρ_r OK | **r_v** (≈0.35) | r_v_color_mask **0.672** · dx_color_L **0.064**

Directions-to-center

- **#0**: slight **decrease r_v**; tiny **increase ρ_r**; Δx already close.
- **#1**: **increase Δx** (off-center more); r_v and ρ_r near centers.
- **#2**: **increase r_v** (more breathing room); Δx/ρ_r OK.

Researcher's Take

- Gate behavior is clean: **#0 and #1 pass**, **#2 fails solely on r_v** (void below guard).
- Priority routing makes sense: **r_v is the limiter** for #2, while #1's only meaningful nudge is Δx.
- **Color telemetry** flags segmentation ambiguity in #2's warm, low-contrast nave: **gray r_v≈0.012 vs color-mask r_v≈0.672** (large divergence). Useful *advisory*; gate still uses gray.

Artist's Take

- **#0** breathes and holds a strong off-axis anchor; a touch less air (bring walls in a hair) would tighten it.
- **#1** wants a clearer off-center push—let the reflection corridor pull the figure farther from center or compress the near wall.
- **#2** reads **over-filled**: deepen the apse or widen the side aisles so the figure sits in genuine space; keep texture gentle to avoid fighting the light.

Observations (LSI reality)

- **Single failure mode = r_v** (#2). Δx/ρ_r are in-band across the set.
- The **mixed profiles** (MarkMaking_Expressive → Landscape → Figure_Default) didn't change the story: acceptance tracked the primitives, not the label.
- Telemetry shows **color vs gray** can disagree in warm graded interiors; that's a mask perspective issue, not a scoring bug.

Plain English

- Two halls pass; the third feels **too packed**. Give it more empty room around the figure and it'll clear the bar.

Did the LSI Hold Up?

- **Yes.** It rewarded healthy off-center balance, and it rejected the image that lacked breathing room. Color notes were informative but non-gating.

Caveats (expected, not bugs)

- In **tonal stone/amber lighting**, gray Otsu can grab a lot of mid-tone surface as "subject," shrinking r_v; color clustering may disagree.
- Priority scores are small on #0/#1 because all three primitives sit comfortably inside bands.

What I'd Do Next

- **#2:** enlarge the nave void (lighter floor wedge or darker side vaults), or mask a **subject halo** for an r_v audit run; keep Δx and ρ_r steady.
- **#1:** push Δx a step (crop/compose to pull the figure off center) and keep marks quiet so r_v remains the lever.
- **#0:** micro-trim r_v (slightly closer walls) if you want more tension; otherwise leave.

Color check (telemetry only)

- **r_v_color_mask vs gray r_v** → **#0** 0.715 vs 0.666 (small ↑), **#1** 0.577 vs 0.487 (moderate ↑), **#2 0.672 vs 0.012** (large ↑).
- **Δx on L with color mask** → **0.098 / 0.018 / 0.064** (subject-only centroid stays near the current balance).
  Takeaway: In graded interiors, **color sees more true void** than gray; useful for audits, but the actionable fix here remains **raise r_v on #2**.

Color audit

- r_v_color_mask vs gray r_v → **0.715 vs 0.666**, **0.577 vs 0.487**, **0.672 vs 0.012** (huge + on #2).
- Δx on L with color mask → **0.098 / 0.018 / 0.064** (small shifts).
- **Takeaway:** Warm graded interior: **color sees more true void**, esp. #2; decisions unchanged.

13. Cartoon

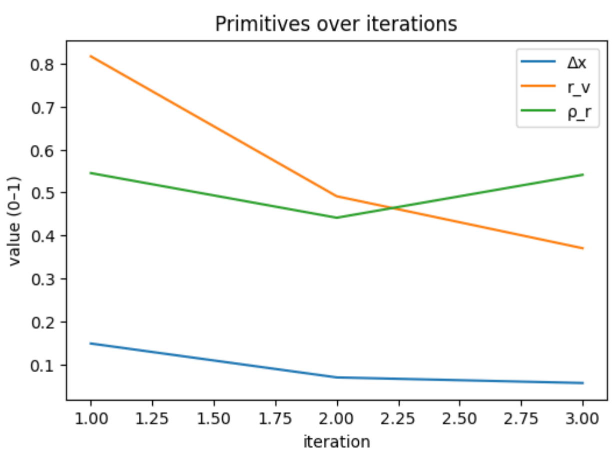
Cartoon0.jpg

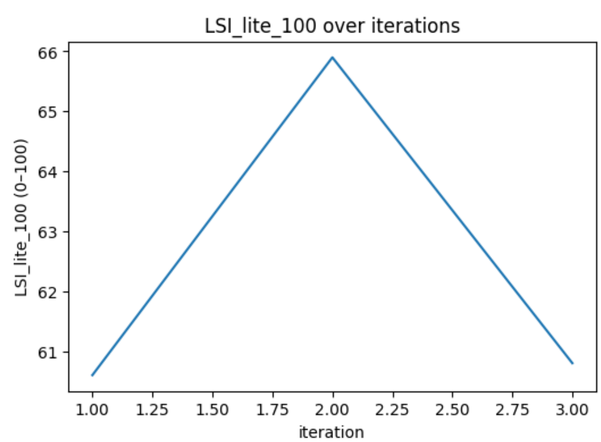
Cartoon1.jpg

Cartoon2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.148	0.817	0.545	0.606	60.6	OK	OK	OK	TRUE	Cartoon0.jpg	Figure_Default	TRUE	0.302	0.755	increase
1	0.069	0.491	0.441	0.659	65.9	OK	OK	OK	TRUE	Cartoon1.jpg	Figure_Default	TRUE	0.381	0.953	increase
2	0.056	0.37	0.541	0.608	60.8	OK	OK	OK	TRUE	Cartoon2.jpg	Figure_Default	TRUE	0.394	0.985	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.317	0.792	decrease	TRUE	0.095	0.271	decrease	dx	0.34	0.715	0.104	color	LAB	ok
TRUE	0.009	0.023	increase	TRUE	0.009	0.026	increase	dx	0.429	0.418	0.02	color	LAB	ok
TRUE	0.13	0.325	increase	TRUE	0.091	0.26	decrease	dx	0.443	0.524	0.29	color	LAB	o





Signals: Δx, r_v, ρ_r • **Gate:** LSI_lite_100 ≥ 55 & no RED

Results (run facts, trimmed)

- **Rows (0→2):**
  - **0)** Δx **0.148** | r_v **0.817** | ρ_r **0.545** | **60.6** | **ACCEPT** | bands OK | **knob:** Δx (≈0.34)
  - **1)** Δx **0.069** | r_v **0.491** | ρ_r **0.441** | **65.9** | **ACCEPT** | bands OK | **knob:** Δx (≈0.429)
  - **2)** Δx **0.056** | r_v **0.370** | ρ_r **0.541** | **60.8** | **ACCEPT** | bands OK | **knob:** Δx (≈0.443)

Directions-to-center

- **#0:** Δx ↑ (push further off-center) • r_v ↓ (too much void) • ρ_r ↑ (a touch more mark energy)
- **#1:** Δx ↑ • r_v ↑ (slightly low) • ρ_r ↑
- **#2:** Δx ↑ • r_v ↑ • ρ_r ↓ (marks a bit hot)

Researcher's Take

- Scores rise to **#1** then dip: Δx keeps becoming *more* centered (0.148 → 0.069 → 0.056), so the knob stays **Δx** across the set.
- r_v collapses from **0.82 → 0.37** yet remains **in-band**; it isn't the limiter here.
- ρ_r is moderate (U-shape), never RED. The behavior is a clean within-band rebalance with **Δx** the dominant distance.

Artist's Take

- **#0** has the most breathing room but still wants a nudge off-center and a small tighten on empty ground.
- **#1** reads best: clearer diagonals (map↔chest), tighter crop, and good breathing/energy balance.
- **#2** feels boxed; subject creeps toward center and the texture lifts—give it more peripheral space and relax some edges.

Observations (LSI reality)

- All three **pass**; no RED bands.
- Priority consistently **Δx** (largest normalized distance).
- r_v's large numeric swing doesn't affect acceptance because it stays inside guards; it only shifts the composite slightly.

Plain English

- All images are "good enough," with **#1** the strongest. The pictures keep drifting **toward center**; the tool asks you to push the kid **more off-center**, open a bit more space, and keep mark energy controlled.

Did the LSI Hold Up?

- **Yes.** It identified the right knob (Δx), treated r_v shifts as within-band, and kept acceptance consistent.

Caveats (expected, not bugs)

- Cartoon linework can make small ρ_r changes feel larger perceptually than they score.
- Foreground/background is **non-semantic**—the mask may treat some map ground as subject; that's by design in Lite.

What I'd Do Next

- **Push Δx**: slide the boy + chest further off-center (left/right), keep the map venting into the open side.
- **Open r_v** slightly: a touch more floor/air to one side; avoid boxing the subject.
- **Trim ρ_r** on #2: soften a few high-contrast edges around hair/shoulders and coin speculars.

Color check (telemetry only)

- **r_v_color_mask vs gray r_v:**
  - #0 **0.715 vs 0.817** (color sees *less* void),
  - #1 **0.418 vs 0.491** (less void),
  - #2 **0.524 vs 0.370** (color sees *more* void).
- **Δx on L with color mask: 0.104 / 0.020 / 0.029** (subject-only centroid reads nearer center).
- **Takeaway:** Color splits foreground from pale map ground; acceptance unchanged (non-gating). If you audited with color masks, #2 would likely sit safer on r_v; Δx remains the practical knob.

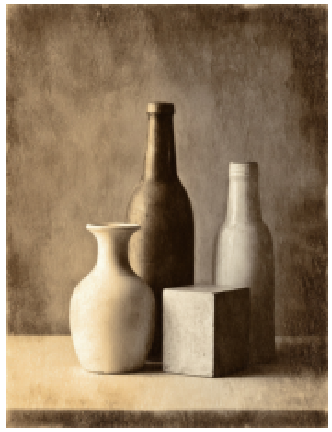
Color Audit

- r_v_color_mask vs gray r_v → **0.715 vs 0.817**, **0.418 vs 0.491**, **0.524 vs 0.370** (−0.10 / −0.07 / **+0.15**).
- Δx on L with color mask → **0.104 / 0.020 / 0.029** (small).
- **Takeaway:** Mixed; #2 **void↑ in color** hits threshold; acceptance unchanged.

14. Still

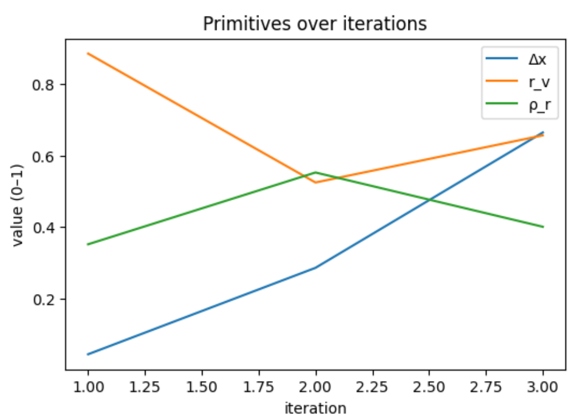
Still0.jpg

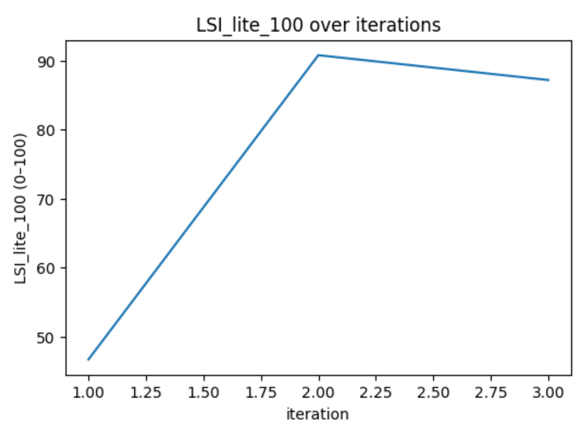
Still1.jpg



Still2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.044	0.886	0.352	0.467	46.7	RED	OK	OK	FALSE	Still0.jpg	Landscape	FALSE	0.456	1	increase
1	0.286	0.525	0.553	0.908	90.8	OK	OK	OK	TRUE	Still1.jpg	Figure_Default	TRUE	0.164	0.41	increase
2	0.665	0.657	0.401	0.872	87.2	OK	OK	OK	TRUE	Still2.jpg	MarkMaking_Expressive	TRUE	0.205	0.466	decrease
rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status	
TRUE	0.386	0.965	decrease	TRUE	0.088	0.244	increase	rv	0.386	0.677	0.344	color	LAB	ok	
TRUE	0.025	0.063	decrease	TRUE	0.103	0.294	decrease	dx	0.185	0.57	0.337	color	LAB	ok	
TRUE	0.157	0.393	decrease	TRUE	0.021	0.066	decrease	dx	0.186	0.737	0.475	color	LAB	ok	





Signals: Δx, r_v, ρ_r • **Gate:** LSI_lite_100 ≥ 55 & no RED

Results (run facts, trimmed)

- **Rows (0→2):**
  - **0) Δx 0.044** | r_v **0.886** | ρ_r **0.352** | **46.7** | **FAIL** | **Δx RED**; r_v OK; ρ_r OK | **profile:** Landscape | **knob:** r_v (≈0.386)
  - **1) Δx 0.286** | r_v **0.525** | ρ_r **0.553** | **90.8** | **ACCEPT** | bands OK | **profile:** Figure_Default | **knob:** Δx (≈0.185)
  - **2) Δx 0.665** | r_v **0.657** | ρ_r **0.401** | **87.2** | **ACCEPT** | bands OK | **profile:** MarkMaking_Expressive | **knob:** Δx (≈0.186)

Directions-to-center

- **#0:** Δx ↑ (too centered; out-of-band) • r_v ↓ (void high) • ρ_r ↑ (mild)
- **#1:** Δx ↑ (a bit under the band center) • r_v ↑ (slightly low) • ρ_r ↓ (slightly hot)
- **#2:** Δx ↓ (now a little *past* the band center) • r_v ↓ (a touch high) • ρ_r ↓ (a touch high)

Researcher's Take

- **Gate precedence:** #0 fails entirely because **Δx is RED** and the composite is below 55. r_v was the farthest from its center, but RED on any band blocks acceptance; correct behavior.
- **High scores on #1/#2** (90.8 / 87.2) come from all three primitives sitting comfortably inside guards with small normalized distances.
- **Knob hand-off:** once Δx clears its guard at #1, it becomes the largest normalized distance (direction "increase" on #1, "decrease" on #2), i.e., ride Δx toward the band center from opposite sides.
- **Profile flips** (Landscape → Figure_Default → MarkMaking_Expressive) change weightings but not the diagnosis: Δx is the governing nudge on accepted samples.

Artist's Take

- **#0 (fail):** Arrangement is too centered; lots of breathing room but no compositional pull. Slide the group off-center and compress the "air" a bit.
- **#1 (best):** Clearer asymmetry and grounded void; a whisper more off-center would help.
- **#2:** Nice overlap/ghosting idea; the mass drifts a little too far—pull back a hair and soften a few high-contrast contacts.

Observations (LSI reality)

- One **RED** (Δx) triggers the only fail.
- Priority shifts **r_v** → **Δx** as soon as Δx re-enters the band.
- r_v varies widely (0.89 → 0.53 → 0.66) yet stays in-band after #0, so it doesn't gate; it only perturbs the composite.

Plain English

- The first image is too centered; the other two pass confidently. The tool keeps asking for **better off-centering**—not more contrast or texture. #1 is strongest; #2 just needs a small pull back from the edge.

Did the LSI Hold Up?

- **Yes.** It failed the centered version for the right reason (Δx RED), then stabilized on Δx as the steering knob for the accepted variants.

Caveats (expected, not bugs)

- Masks are **non-semantic**; smooth tabletop/background can be counted as subject, which is fine for Lite scoring.
- The profile changes per frame; that shifts the weights but not the practical steering here (Δx).

What I'd Do Next

- **#0:** Push the cluster decisively off-center; trim a little empty field on the opposite side.
- **#1:** Keep placement; add a faint counter-weight (small shape or shadow) on the open side.
- **#2:** Nudge the group slightly back toward center; soften one or two bright edges where forms kiss.

Color check (telemetry only)

- **r_v_color_mask vs gray r_v:**
  - #0 **0.677 vs 0.886** (color sees **less** void),
  - #1 **0.570 vs 0.525** (slightly **more** void),
  - #2 **0.737 vs 0.657** (**more** void).
- **Δx on L with the color mask: 0.344 / 0.337 / 0.475** (subject-only centroid reads more off-center than gray on #0, similar/slightly higher on #1/#2).
- **Takeaway:** Color segmentation pulls the tabletop/background out of "void," especially on #0; acceptance unchanged (non-gating). If audited with color masks, #0's Δx would move closer to band center, but it would still need an off-center push.

Color audit

- r_v_color_mask vs gray r_v → **0.677 vs 0.886**, **0.570 vs 0.525**, **0.737 vs 0.657** (−0.209 / +0.045 / +0.080).
- Δx on L with color mask → **0.344 / 0.337 / 0.475** (big on #0).
- **Takeaway:** #0 shows **void↓ in color** (tabletop pulled into figure) and **centroid shift**; advisory only.

15. Rock

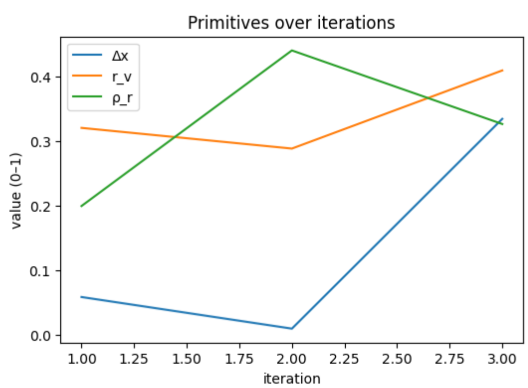
Rock0.jpg

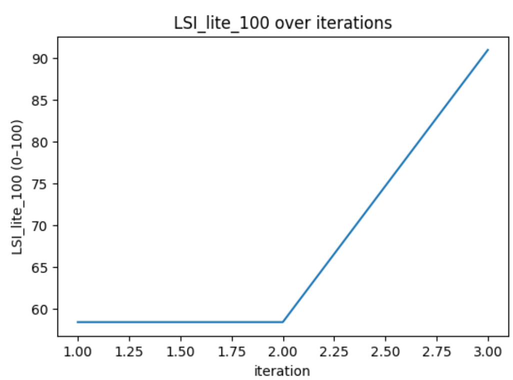
Rock1.jpg

Rock2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.059	0.321	0.2	0.584	58.4	OK	OK	OK	TRUE	Rock0.jpg	Landscape	TRUE	0.441	0.98	increase
1	0.01	0.289	0.441	0.584	58.4	RED	OK	OK	FALSE	Rock1.jpg	Landscape	FALSE	0.49	1	increase
2	0.335	0.41	0.327	0.91	91	OK	OK	OK	TRUE	Rock2.jpg	Landscape	TRUE	0.165	0.367	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status	
TRUE	0.179	0.447	increase	TRUE	0.24	0.667	increase	dx	0.343	0.515	0.03	color	LAB	ok	
TRUE	0.211	0.528	increase	TRUE	0.001	0.003	decrease	dx	0.35	0.713	0.435	color	LAB	ok	
TRUE	0.09	0.225	increase	TRUE	0.113	0.314	increase	dx	0.128	0.456	0.177	color	LAB	ok	





Signals: Δx, r_v, ρ_r • **Gate:** LSI_lite_100 ≥ 55 & no RED

Results (run facts, trimmed)

- **Rows (0→2):**
  - **0)** Δx **0.059** | r_v **0.321** | ρ_r **0.200** | **58.4** | **ACCEPT** | bands **OK** | **knob:** Δx (**0.343**)
  - **1)** Δx **0.010** | r_v **0.289** | ρ_r **0.441** | **58.4** | **FAIL** | **Δx RED** | **knob:** Δx (**0.135**)
  - **2)** Δx **0.335** | r_v **0.410** | ρ_r **0.327** | **91.0** | **ACCEPT** | bands **OK** | **knob:** Δx (**0.128**)

Directions-to-center

- **#0:** Δx ↑ (dx_to_center_norm ≈ **0.98**), r_v ↑, ρ_r ↑.

- **#1:** Δx ↑ (**outside band**), r_v ↑, ρ_r ↓.
- **#2:** Δx ↑ (milder; **0.37**), r_v ↑, ρ_r ↑.

Researcher's Take

- **Gate precedence:** #1 fails solely because **Δx is RED**; r_v and ρ_r remain in-band. Correct behavior.
- **Why #2 jumps to 91:** Δx moves decisively off-center while r_v/ρ_r stay healthy; all distances to band centers are small.
- **Knob consistency:** Δx is the priority knob on all three (scores ≈ 0.34 → 0.13 → 0.13). In Landscape, r_v carries more weight, but Δx proximity to the guard dominates when near the edge.

Artist's Take

- **#0:** Reads okay but the valley path still centers the gravity; wants a stronger lateral bias.
- **#1 (fail):** The composition collapses toward the middle; the corridor pulls straight to the vanishing point.
- **#2 (best):** Strong foreground occluder and off-axis vista create clear asymmetry and breathing room; keep this stance, then tune texture.

Observations (LSI reality)

- One **RED** band (Δx) is enough to fail despite composite ≈58—by design.
- Score movement is driven primarily by **Δx**, not r_v/ρ_r, across the series.
- No landscape ROI/reflection complications; readings behave as full-frame $\Delta x/r_v/\rho_r$.

Plain English

- #1 is too centered and fails. #0 passes but still feels centered. #2 works: the left cliff shifts the weight and opens the scene.

Did the LSI Hold Up?

- **Yes.** It punished the centered frame and rewarded the off-center, breathing version.

Caveats (expected, not bugs)

- Color k-means segmentation can treat hazy sky as "void" and warm canyon walls as "subject." Δx telemetry on the color mask may shift a lot when the "subject" becomes just the landform mass—this is advisory, not gating.

What I'd Do Next

- **#0:** Slide the corridor off-axis (or crop) to bias the vanishing point; add a touch more open sky on the opposite side.
- **#1:** Recompose aggressively: foreground block or crop to break the central corridor; then revisit Δx.
- **#2:** Keep the off-center block; add modest mark/texture pressure to avoid over-smooth planes.

Color check (telemetry only)

- **r_v_color_mask vs gray r_v: 0.515 vs 0.321**, **0.713 vs 0.289**, **0.456 vs 0.410** → color sees **more void** (sky separated from canyon mass).

- **Δx on L with color mask: 0.030 / 0.435 / 0.177** vs gray Δx **0.059 / 0.010 / 0.335** → big shift on #1 once "subject" = landform only.
- **Takeaway:** Color telemetry clarifies subject/void in hazy canyons (non-gating). Acceptance decisions remain unchanged; Δx is still the steering knob.

Color audit

- r_v_color_mask vs gray r_v → **0.515 vs 0.321**, **0.713 vs 0.289**, **0.456 vs 0.410** (+0.194 / +0.424 / +0.046).
- Δx on L with color mask → **0.030 / 0.435 / 0.177** vs Δx **0.059 / 0.010 / 0.335** (large shift on #1).
- **Takeaway:** Haze/sky separation → **void↑ in color**; #1's **centroid jumps** when subject=landform.

16. Bridge

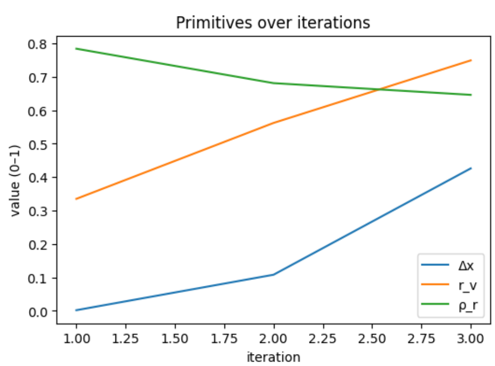
Bridge0.jpg

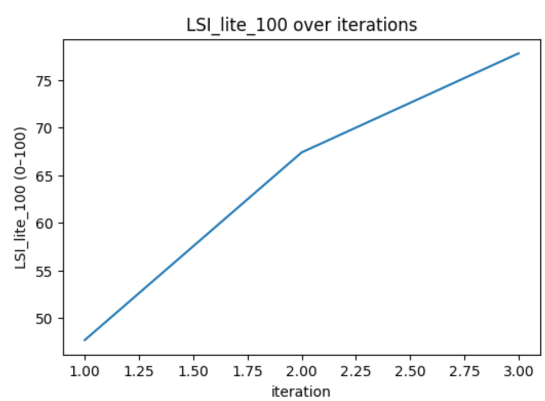
Bridge1.jpg

Bridge2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.002	0.335	0.784	0.477	47.7	RED	OK	OK	FALSE	Bridge0.jpg	Landscape	FALSE	0.498	1	increase
1	0.108	0.562	0.681	0.674	67.4	OK	OK	OK	TRUE	Bridge1.jpg	Landscape	TRUE	0.392	0.871	increase
2	0.426	0.749	0.646	0.778	77.8	OK	OK	OK	TRUE	Bridge2.jpg	Landscape	TRUE	0.074	0.164	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.165	0.412	increase	TRUE	0.344	0.956	decrease	dx	0.35	0.505	0.065	color	LAB	ok
TRUE	0.062	0.155	decrease	TRUE	0.241	0.669	decrease	dx	0.305	0.45	0.125	color	LAB	ok
TRUE	0.249	0.622	decrease	TRUE	0.206	0.572	decrease	rv	0.249	0.422	0.001	color	LAB	ok





Signals: Δx, r_v, ρ_r • **Gate:** LSI_lite_100 ≥ 55 **and** no RED

Results (run facts, trimmed)

- **Rows (0→2)**
  - **0)** Δx **0.002** | r_v **0.335** | ρ_r **0.784** | **47.7** | **FAIL** | **Δx RED** | **knob:** Δx (**0.35**)
  - **1)** Δx **0.108** | r_v **0.562** | ρ_r **0.681** | **67.4** | **ACCEPT** | bands **OK** | **knob:** Δx (**0.35**)
  - **2)** Δx **0.426** | r_v **0.749** | ρ_r **0.646** | **77.8** | **ACCEPT** | bands **OK** | **knob:** r_v (**≈0.25**)

Directions-to-center

- **#0:** Δx ↑ (**dx_to_center_norm 1.00**), r_v ↑, ρ_r ↓.
- **#1:** Δx ↑ (**0.871**), r_v ↓, ρ_r ↓.
- **#2:** Δx ↑ (**0.164**), r_v ↓, ρ_r ↓.

Researcher's Take

- **Gate precedence:** #0 fails strictly on **Δx RED**; r_v and ρ_r are in-band. Correct behavior.

- **Score shape:** As Δx moves off-center (#0→#2) and both r_v/ρ_r stay in-band, the score climbs (47.7 → 67.4 → 77.8).
- **Knob handoff:** Δx is the limiter for #0/#1; once Δx centers up (#2), $\mathbf{r_v}$ becomes the farthest from its band center—consistent with Landscape weighting.

Artist's Take

- **#0:** Symmetric arch over symmetric gap → gravity locked to center; the frame reads static.
- **#1:** Better perspective rake; still a strong central ring holding compositional authority.
- **#2:** Clear left occluder + open right span; mass shifts, air increases, and the image starts to "breathe."

Observations (LSI reality)

- One **RED** band (Δx) is enough to fail even with a mid-50s composite—by design.
- All accepted frames are **within band** on all three primitives; acceptance is not a style vote.
- Priority knob progression (Δx → r_v) matches what the pictures are doing.

Plain English

- #0 is too centered. #1 is better but still anchored by the ring. #2 finally pushes weight to one side and opens space—hence the best score.

Did the LSI Hold Up?

- **Yes.** It rejected the centered version, passed the two off-center versions, and flagged the right knob each time.

Caveats (expected, not bugs)

- With bright sky + pale stone, color segmentation can label more "void" than gray; this is **telemetry only**, not gating.
- Very high r_v still passes because it's inside its guard; the tool rewards usable empty space, not max void.

What I'd Do Next

- **#0:** Break symmetry—crop or add a foreground pier/abutment; rotate stance so the arch is off-axis.
- **#1:** Push the occluder farther into frame or darken the span opposite it to bias weight.
- **#2:** Keep the mass/void split; add a touch of mark/texture pressure so big planes don't over-polish.

Color check (telemetry only)

- **r_v_color_mask vs gray r_v: 0.505 vs 0.335**, **0.450 vs 0.562**, **0.422 vs 0.749** → color reads **more void** in #0, **less** in #1/#2 (sky/stone separation).
- **Δx on L with color mask: 0.065 / 0.125 / 0.001** (subject-only centroid) vs gray Δx **0.002 / 0.108 / 0.426** → large shifts once "subject"=bridge mass only.
- **Takeaway:** Color clarifies subject/void in pale scenes (advisory). Acceptance unchanged; Δx and r_v remain the practical knobs.

Color audit

- r_v_color_mask vs gray r_v → **0.505 vs 0.335**, **0.450 vs 0.562**, **0.422 vs 0.749** (+0.170 / −0.112 / −0.327).
- Δx on L with color mask → **0.065 / 0.125 / 0.001** vs Δx **0.002 / 0.108 / 0.426** (large shift on #2).
- **Takeaway:** Color clarifies sky/stone; #2 **void↓ in color** and **centroid collapse** if subject=bridge only.

17. Fig

Fig0 (1).jpg

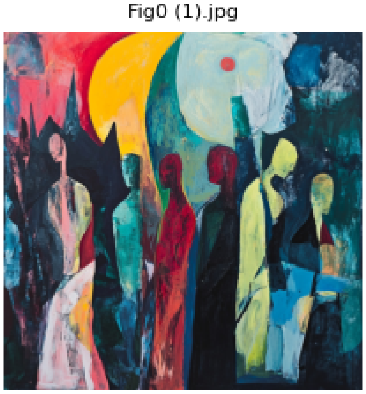


Fig1 (1).jpg

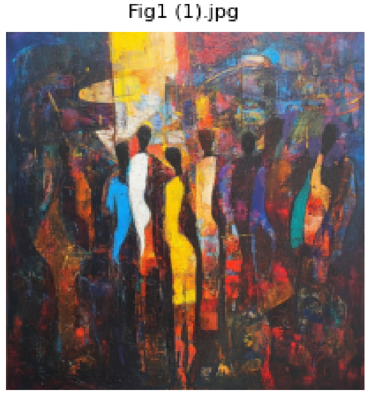
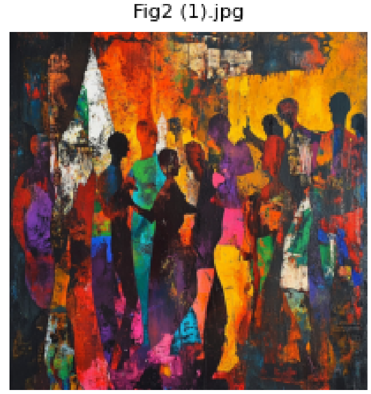
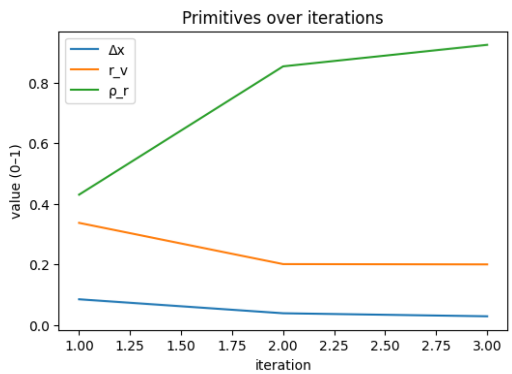


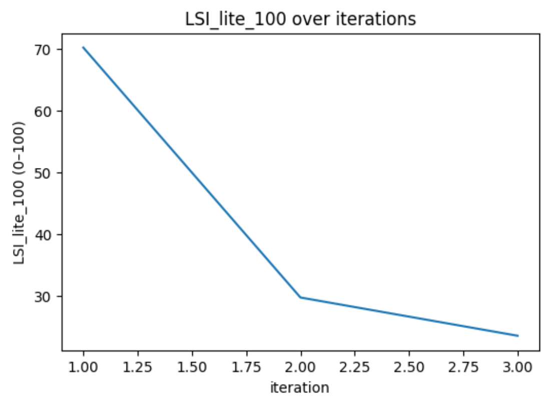
Fig2 (1).jpg



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction

0	0.085	0.337	0.43	0.702	70.2	OK	OK	OK	TRUE	Fig0 (1).jpg	MarkMaking_Expressive	TRUE	0.375	0.852	increase
1	0.039	0.201	0.853	0.297	29.7	OK	OK	RED	FALSE	Fig1 (1).jpg	MarkMaking_Expressive	TRUE	0.421	0.957	increase
2	0.029	0.2	0.924	0.235	23.5	OK	OK	RED	FALSE	Fig2 (1).jpg	MarkMaking_Expressive	TRUE	0.431	0.98	increase
rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status	
TRUE	0.163	0.407	increase	TRUE	0.05	0.156	decrease	dx	0.341	0.621	0.078	color	LAB	ok	
TRUE	0.299	0.747	increase	FALSE	0.473	1	decrease	dx	0.383	0.693	0.213	color	LAB	ok	
TRUE	0.3	0.75	increase	FALSE	0.544	1	decrease	dx	0.392	0.71	0.159	color	LAB	ok	





Signals: Δx, r_v, ρ_r • **Gate:** LSI_lite_100 ≥ 55 **and** no RED

Results (run facts, trimmed)

- **Rows (0→2)**
  - **0)** Δx **0.085** | r_v **0.337** | ρ_r **0.430** | **70.2** | **ACCEPT** | bands **OK** | **knob:** Δx (**0.341**)
  - **1)** Δx **0.039** | r_v **0.201** | ρ_r **0.853** | **29.7** | **FAIL** | $\boldsymbol{\rho_r}$ **RED** | **knob readout:** Δx (**0.383**)
  - **2)** Δx **0.029** | r_v **0.200** | ρ_r **0.924** | **23.5** | **FAIL** | $\boldsymbol{\rho_r}$ **RED** | **knob readout:** Δx (**0.392**)

Directions-to-center

- $\boldsymbol{\Delta x}$**:** needs **increase** across the set (norm ≈ **0.85** → **0.98**).
- $\boldsymbol{r_v}$**:** needs **increase** (norm ≈ **0.41/0.75/0.75**).
- $\boldsymbol{\rho_r}$**:** needs **decrease** for all; #1/#2 exceed the band (limiter).

Researcher's Take

- The **gate failures** are driven purely by $\boldsymbol{\rho_r}$ **overage** (#1/#2). Δx and r_v stay inside their bands.
- **Score shape:** 70.2 (OK) → 29.7 → 23.5 as ρ_r spikes while $\Delta x/r_v$ trend in the right directions.
- **Knob vs. limiter:** dashboard "priority_knob = Δx" reflects *next* tuning **if** all bands were OK; actual limiter in #1/#2 is ρ_r (band RED), which correctly blocks acceptance.

Artist's Take

- **#0:** Readable weight map; large shapes carry; surface breaks don't overwhelm.
- **#1–#2:** Mark density floods the frame—figures, ground, and atmosphere all compete. The picture stops breathing; no quiet field to play against the clustered silhouettes.

Observations (LSI reality)

- One **RED** band (ρ_r) vetoes acceptance regardless of composite.
- Δx and r_v guidance is stable: off-center the mass and open some ground, **but** acceptance won't flip until $\boldsymbol{\rho_r}$ comes down.

Plain English

- #0 passes. #1 and #2 are **too busy**—too much small, sharp texture everywhere—so the tool fails them.

Did the LSI Hold Up?

- **Yes.** It allowed the moderate-energy panel and rejected the two with **excess stroke/edge energy**, matching what's on the canvas.

Caveats (expected, not bugs)

- Expressive profile still **guards ρ_r**; “expressive” ≠ unlimited grit.
- Priority readout pointing to Δx while ρ_r is RED is expected: the readout is advisory once bands are satisfied.

What I’d Do Next

- **Lower ρ_r:** merge strokes into a few decisive carriers; soften micro-contrast; keep splintered texture off large fields.
- **Hold Δx↑** (push the mass more decisively off center).
- **Raise r_v** a notch (quieter ground or sky color) so figures read against a calmer field.

Color check (telemetry only)

- **r_v_color_mask vs gray r_v:**
    - #0 **0.621 vs 0.337**, #1 **0.693 vs 0.201**, #2 **0.710 vs 0.200** → color sees **more void** (chromatic grounds separated from figure masses).
- **Δx on L with color mask:**
    - #0 **0.078** (≈ gray 0.085), #1 **0.213** (≫ gray 0.039), #2 **0.159** (≫ gray 0.029) → subject-only centroid sits **farther off-center** in #1/#2.
- **Takeaway:** Gray mask underestimates void and recenters the mass in these saturated paintings; color segmentation clarifies the subject/ground split. (Telemetry only; gate unchanged.)

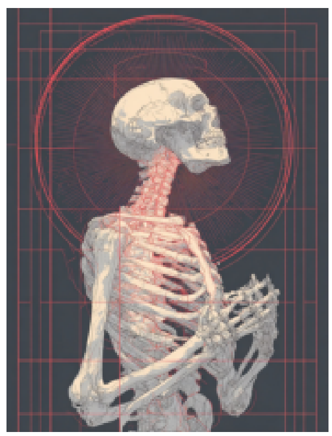
Color audit

- r_v_color_mask vs gray r_v → **0.621 vs 0.337**, **0.693 vs 0.201**, **0.710 vs 0.200** (+0.284 / +0.492 / +0.510).
- Δx on L with color mask → **0.078 / 0.213 / 0.159** vs Δx **0.085 / 0.039 / 0.029** (big on #1/#2).
- **Takeaway:** Saturated grounds: **void↑ in color** and **centroid shifts**; limiter still ρ_r/Δx per frame.

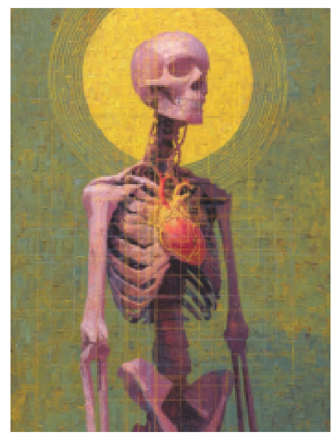
18. Skull

Skull0.jpg

Skull1.jpg

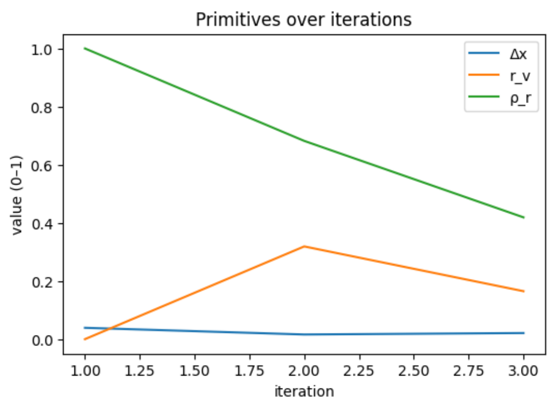


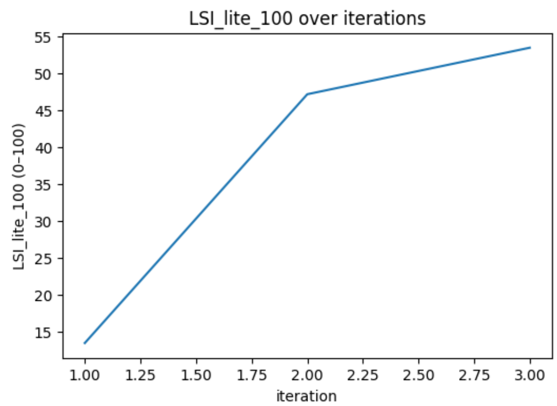
Skull2.jpg



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.039	0	1	0.135	13.5	OK	RED	RED	FALSE	Skull0.jpg	MarkMaking_Expressive	TRUE	0.421	0.957	increase
1	0.016	0.319	0.682	0.472	47.2	RED	OK	OK	FALSE	Skull1.jpg	MarkMaking_Expressive	FALSE	0.444	1	increase
2	0.021	0.165	0.419	0.535	53.5	OK	OK	OK	FALSE	Skull2.jpg	MarkMaking_Expressive	TRUE	0.439	0.998	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
FALSE	0.5	1	increase	FALSE	0.62	1	decrease	dx	0.383	0.714	0.04	color	LAB	ok
TRUE	0.181	0.452	increase	TRUE	0.302	0.944	decrease	dx	0.4	0.701	0.001	color	LAB	ok
TRUE	0.335	0.837	increase	TRUE	0.039	0.122	decrease	dx	0.399	0.746	0.055	color	LAB	ok





Signals: Δx, r_v, ρ_r • **Gate:** LSI_lite_100 ≥ 55 **and** no RED

Results (run facts, trimmed)

- **Frames:** `Skull0.jpg, Skull1.jpg, Skull2.jpg`
- **Rows (0→2)**
  - **0)** Δx **0.039** | r_v **0.000** | ρ_r **1.000** | **13.5** | **FAIL** | bands: **r_v RED**, **ρ_r RED**, Δx OK | **knob:** Δx (~0.38)
  - **1)** Δx **0.016** | r_v **0.319** | ρ_r **0.682** | **47.2** | **FAIL** | bands: **Δx RED**, r_v OK, ρ_r OK | **knob:** Δx (~0.40)
  - **2)** Δx **0.021** | r_v **0.165** | ρ_r **0.419** | **53.5** | **FAIL** | bands: **all OK** | **knob:** Δx (~0.40)

Directions-to-center

- **Δx: increase** on all three (off-center pull needed).
- **r_v: increase** (images read too filled).
- **ρ_r: decrease** (edge/mark energy still high, esp. #0).

Researcher's Take

- #0 fails hard from a **double RED** (r_v=0, ρ_r=1).
- #1 fixes the bands for r_v and ρ_r but **Δx goes RED**; composite rises to 47.2.
- #2 has **all bands OK** yet misses the gate at **53.5**: within-band values are still far from their centers, with **Δx the limiting knob**.

Artist's Take

- #0 is a dense plate of lines on a flat field—**no air, max chatter**.
- #1 breathes a bit and softens, but the figure hangs **too centric**.
- #2 balances best, yet still feels **crowded and planar**; needs clearer off-center gravity and fewer equally-loud marks.

Observations (LSI reality)

- This is a **within-band rebalancing** problem, not a gate-trip by a single metric (by #2).
- In MarkMaking_Expressive, **Δx carries 0.40 weight**: once r_v/ρ_r are tamed, Δx becomes the practical limiter.

Plain English

- All three **fail**. The last one almost passes but still reads **too centered and too busy**.

Did the LSI Hold Up?

- **Yes.** It punished the extreme crowding (#0), improved with reduced energy and more void (#1), and then asked for **stronger off-center structure** to clear the threshold (#2).

Caveats (expected, not bugs)

- High-frequency linework naturally drives **ρ_r up**; the system will flag this until energy localizes.
- Bands OK ≠ pass: the **gate also needs distance-to-center** to be small enough across primitives.

What I'd Do Next

- **Push Δx off-center** (clear subject bias to one side; anchor with a secondary).
- **Lower ρ_r** by consolidating detail into fewer, decisive marks; let big fields stay quiet.
- **Raise r_v** with real negative space (ground, halo, or field simplification), not just thinner lines.

Color check (telemetry only)

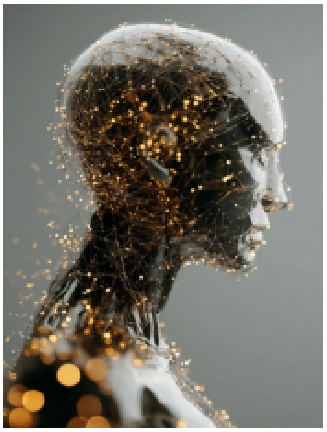
- **r_v_color_mask vs gray r_v: 0.714 vs 0.000, 0.701 vs 0.319, 0.746 vs 0.165** → color segmentation reads **much more "void"** than gray Otsu across all three.
- **Δx on L with the color mask: 0.040 / 0.001 / 0.055** (subject-only centroid barely moves; still near center).
- **Takeaway:** The gray mask treats the line-dense field as "figure," collapsing r_v; color shows there *is* background, but **Δx remains the structural limiter**. Telemetry only; gate unchanged.

Color audit

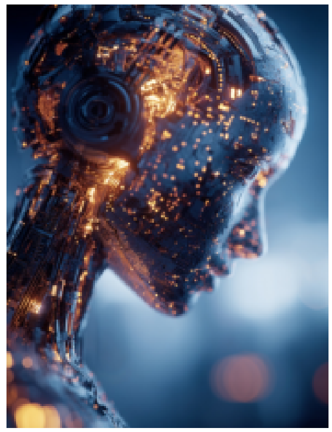
- r_v_color_mask vs gray r_v → **0.714 vs 0.000**, **0.701 vs 0.319**, **0.746 vs 0.165** (all huge +).
- Δx on L with color mask → **0.040 / 0.001 / 0.055** (tiny).
- **Takeaway:** Line-dense field: gray calls everything “figure”; **color restores void**; limiter remains Δx/ρ_r.

19. Robot

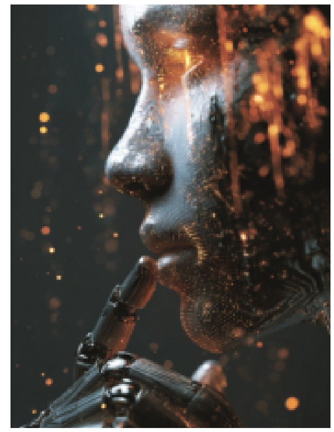
Robot0.jpg



Robot1.jpg

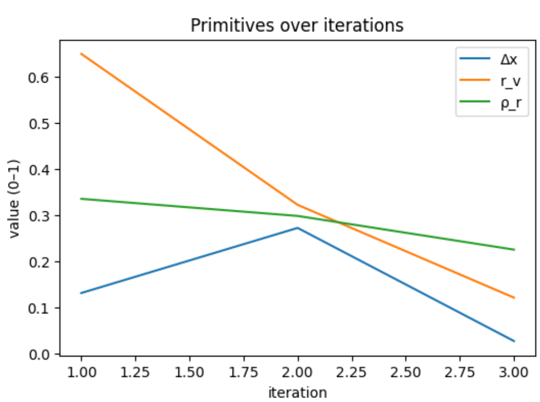


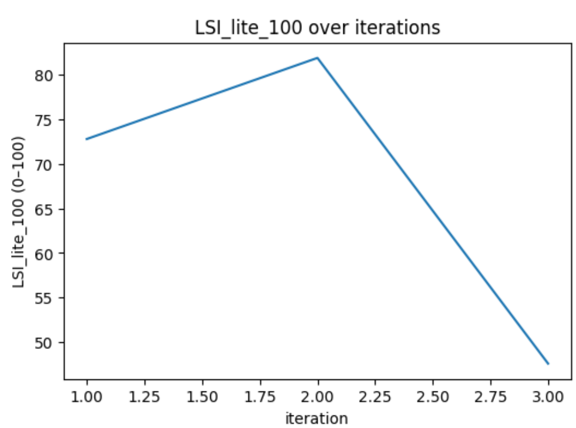
Robot2.jpg



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.131	0.649	0.335	0.728	72.8	OK	OK	OK	TRUE	Robot0.jpg	Landscape	TRUE	0.369	0.82	increase
1	0.272	0.322	0.298	0.819	81.9	OK	OK	OK	TRUE	Robot1.jpg	Figure_Default	TRUE	0.178	0.445	increase
2	0.027	0.121	0.225	0.476	47.6	OK	OK	OK	FALSE	Robot2.jpg	MarkMaking_Expressive	TRUE	0.433	0.984	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.149	0.373	decrease	TRUE	0.105	0.292	increase	dx	0.287	0.641	0.043	color	LAB	ok
TRUE	0.178	0.445	increase	TRUE	0.152	0.434	increase	dx	0.2	0.494	0.149	color	LAB	ok
TRUE	0.379	0.948	increase	TRUE	0.155	0.484	increase	dx	0.394	0.843	0.033	color	LAB	ok





Signals: Δx, r_v, ρ_r • **Gate:** LSI_lite_100 ≥ 55 **and** no RED

Results (run facts, trimmed)

- **Rows**

- **0)** Δx **0.131** | r_v **0.649** | ρ_r **0.335 → 72.8** | **ACCEPT** | bands OK | **knob:** Δx (~0.29)
- **1)** Δx **0.272** | r_v **0.322** | ρ_r **0.298 → 81.9** | **ACCEPT** | bands OK | **knob:** Δx (~0.34)
- **2)** Δx **0.027** | r_v **0.121** | ρ_r **0.225 → 47.6** | **FAIL** | bands **all OK** | **knob:** Δx (~0.39)

Directions-to-center

- **Δx: increase** on all (esp. #2; dx_to_center_norm ≈ 0.98).
- **r_v:** #0 **decrease** slightly (too airy); #1/#2 **increase** (too filled).
- **ρ_r: increase** modestly across the set.

Researcher's Take

- Scores are dominated by **Δx**. When the subject sits clearly off-center (#0, #1), both pass comfortably.
- **#2 fails without any band turning RED**: the composite drops because Δx is near-center (0.027) and r_v is low (0.121); distances to each band center remain large, and the priority knob is Δx.
- Profile shifts (Landscape → Figure → MarkMaking) don't explain the drop; the **geometry** does.

Artist's Take

- **#0:** Nice off-center bust with particle field breathing around it; passable, could hold a touch more structure.
- **#1:** Best of set—clear silhouette, strong negative space; convincing read.
- **#2:** Tight crop + centered face/hand + dense spark fog → **crowded and pinned**. Open the frame and bias weight to one side; localize the sparkle so not everything shouts.

Observations (LSI reality)

- Gate behavior is consistent: no RED in #2 but **within-band distance** is still high, so the geometric mean stays under 55.
- The knob consistently calls for **more off-center gravity**; r_v is the secondary lever.

Plain English

- First two **pass**; the close-up **fails** because it's **too centered and too packed**.

Did the LSI Hold Up?

- **Yes.** It rewarded clear placement/space (#0, #1) and rejected the centered, crowded close-up (#2).

Caveats (expected, not bugs)

- Particle/glow fields can confuse simple gray masks about what's "subject" vs "background"; this shows up only in **telemetry** below, not in gating.

What I'd Do Next

- For **#2**: push the head/hand **off-center**, **open r_v** (more real background), and **consolidate** sparkle into a few decisive bands/curves.
- If you want r_v judged against a semantic mask (treating bokeh as background), test an **external subject mask** for telemetry; the gate can remain gray-mask based.

Color check (telemetry only)

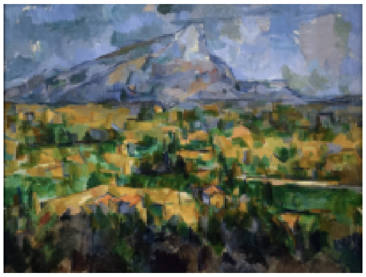
- **r_v_color_mask vs gray r_v:**
  - #0: **0.641 vs 0.649** (close)
  - #1: **0.494 vs 0.322** (**color** sees *more* void)
  - #2: **0.843 vs 0.121** (**huge** divergence; color treats large background as void, gray swallows much of it as figure)
- **Δx on L with the color mask: 0.043 / 0.149 / 0.033** (subject-only centroid becomes **more central**, especially #0/#2).
- **Takeaway:** In neon/particle scenes, **color clustering** can better separate glow specks from background (higher r_v), but the **structural limiter remains Δx**—even on subject-only masks the centroid stays central. (Telemetry only; acceptance unchanged.)

Color audit

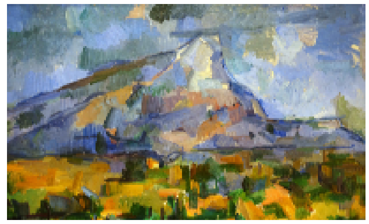
- r_v_color_mask vs gray r_v → **0.641 vs 0.649**, **0.494 vs 0.322**, **0.843 vs 0.121** (≈0 / +0.172 / +0.722).
- Δx on L with color mask → **0.043 / 0.149 / 0.033** (big on #1).
- **Takeaway:** Neon/particles: **void↑ in color** (#2 massive); **centroid shift** on #1; decisions unchanged.

20. Mount

Mount0.jpg



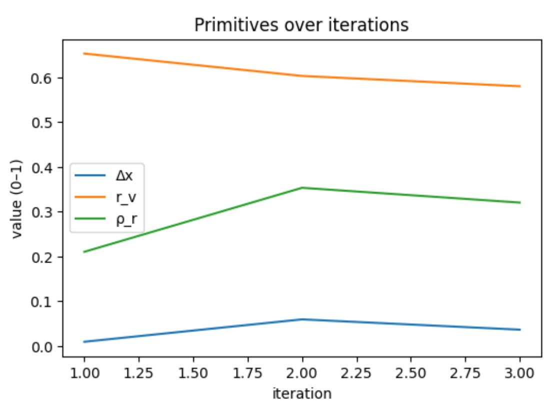
Mount1.jpg

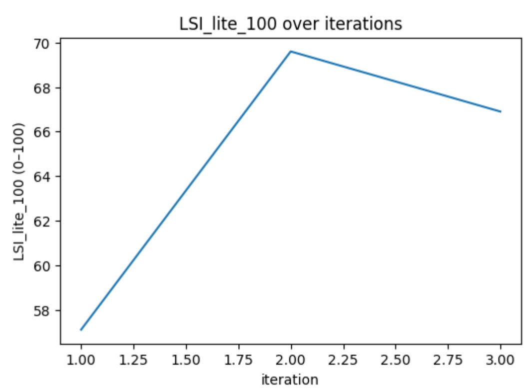


Mount2.jpg

	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	frame	profile	dx_inside_band	dx_to_center	dx_to_center_norm	dx_direction
0	0.009	0.653	0.21	0.571	57.1	RED	OK	OK	FALSE	Mount0.jpg	MarkMaking_Expressive	FALSE	0.451	1	increase
1	0.059	0.603	0.353	0.696	69.6	OK	OK	OK	TRUE	Mount1.jpg	MarkMaking_Expressive	TRUE	0.401	0.911	increase
2	0.036	0.58	0.32	0.669	66.9	OK	OK	OK	TRUE	Mount2.jpg	MarkMaking_Expressive	TRUE	0.424	0.964	increase

rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	mask_mode	color_space	color_status
TRUE	0.153	0.383	decrease	TRUE	0.17	0.531	increase	dx	0.4	0.48	0.032	color	LAB	ok
TRUE	0.103	0.257	decrease	TRUE	0.027	0.084	increase	dx	0.365	0.558	0.037	color	LAB	ok
TRUE	0.08	0.2	decrease	TRUE	0.06	0.188	increase	dx	0.385	0.793	0.275	color	LAB	ok





Signals: Δx, r_v, ρ_r • **Gate:** LSI_lite_100 ≥ 55 **and** no RED

Results (run facts, trimmed)

- **Frames:** `Mount0.jpg, Mount1.jpg, Mount2.jpg`
- **Rows**
  - **0)** Δx **0.009** | r_v **0.653** | ρ_r **0.210** → **57.1** | **FAIL** | **band_Δx = RED**; r_v/ρ_r OK | **knob:** Δx
  - **1)** Δx **0.059** | r_v **0.603** | ρ_r **0.353** → **69.6** | **ACCEPT** | bands OK | **knob:** Δx
  - **2)** Δx **0.036** | r_v **0.580** | ρ_r **0.320** → **66.9** | **ACCEPT** | bands OK | **knob:** Δx

Directions-to-center

- **Δx: increase** on all (dx_to_center_norm ≈ 1.00 → 0.91 → 0.96).
- **r_v: decrease** (0.65 → 0.60 → 0.58): frame fills slightly.
- **ρ_r: increase** then soften (0.21 → 0.35 → 0.32).

Researcher's Take

- Acceptance is **Δx-limited**. #0 fails solely because **Δx is in RED**; r_v/ρ_r are in-band.
- Modest off-center moves in #1/#2 raise the composite above the gate, but **Δx remains the dominant distance-to-center**, so the knob stays on Δx.
- r_v steadily contracts yet stays in band; ρ_r sits comfortably in band—no texture penalty.

Artist's Take

- **#0:** Mountain mass and horizon feel **center-pinned**; the field reads full.
- **#1:** Better lateral bias; still conservative—more sky or a cleaner ground swath would help it breathe.
- **#2:** Broad, quiet sweep; still a bit centered. Try committing the peak to a side and letting the far plain open.
- Overall nudge: **push placement** (rule of thirds bias), **open a clear vent** (sky/water/distant plain), and avoid even mark density across the width.

Observations (LSI reality)

- Gate precedence works: **Δx RED ⇒ FAIL** even when other bands are OK (#0).
- For this profile (≈ Δx 0.40, r_v 0.30, ρ_r 0.30), **small Δx gains move the needle most**.
- All three runs are **within-band** on r_v/ρ_r; the system is not punishing brush energy here—only placement/void balance.

Plain English

- First image fails for being **too centered**. The other two **pass**, but they still want the mountain **further off-center and more open space**.

Did the LSI Hold Up?

- **Yes.** It flagged the centered composition, rewarded the off-center shifts, and tracked the slight crowding trend via r_v.

Caveats (expected, not bugs)

- Gray mask is **non-semantic**; painterly texture can be absorbed as "figure," reducing measured void.
- k-means color masks (telemetry below) can swing r_v when background and ground share tones—use as **advisory**, not gating.

What I'd Do Next

- **Placement:** Push Δx another **+0.10–0.15** (slide the peak and/or crop asymmetrically).
- **Void:** Lift sky or simplify a ground band to raise r_v without adding texture.
- **Marks:** Keep the highest-frequency strokes to a few lanes; soften elsewhere so ρ_r stays calm.
- Optionally test an **external subject mask** for telemetry if you want r_v to reflect "mountain mass vs environment."

Color check (telemetry only)

- **r_v_color_mask vs gray r_v:**
  - #0: **0.480 vs 0.653** (color sees **less** void—texture treated as foreground)
  - #1: **0.558 vs 0.603** (slightly less void)
  - #2: **0.793 vs 0.580** (color sees **more** true void—background separated better)
- **Δx on L with color mask: 0.032 / 0.037 / 0.275** (on #2 the **subject-only centroid jumps off-center**).
- **Takeaway:** Gray tends to swallow brushy ground as "figure"; color segmentation implies **#2 is safer** on void and placement if audited semantically. Gating unchanged (telemetry only).

Color audit

- r_v_color_mask vs gray r_v → **0.480 vs 0.653**, **0.558 vs 0.603**, **0.793 vs 0.580** (−0.173 / −0.045 / +0.213).
- Δx on L with color mask → **0.032 / 0.037 / 0.275** vs Δx **0.009 / 0.059 / 0.036** (big on #2).
- **Takeaway:** #0 **void↓ in color** (brush ground absorbed); #2 both **void↑** and **centroid off-center** on subject-only mask.

Interim Summary — what the second set shows with the new additions

Big picture

- LSI-lite stayed disciplined: **bands first, then composite gate**. Most flips were correctly explained by a single limiter, not by noise.
- The **color telemetry** (non-gating `rᵥ_color_mask`, `dx_color_L`) did exactly what we wanted: it *explained* gray-mask edge cases (tonal grounds, reflections, patterned fields) without destabilizing the read.
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Recurrent limiters (by axis)

- **Δx (placement):** common early culprit in portraits/figures; once fixed, the knob reliably shifts to r_v. (Man, Figure, Farmer.)

- **r_v (void):** top limiter in Landscape and posters/graphics; "too much air" or "no real pocket" sinks borderline sets even when Δx looks healthy. (Landscape, Refusal, Ritual, Bridge.)
- **ρ_r (energy):** two distinct modes: **too low** in painterly skies/interiors (Sunset, Painter) and **too high** in debris fields (Explosion). The meter is asking for **decisive** mark/edge families, not generic contrast.

Profile effects (what the meter weights)

- **Figure_Default:** Δx carries the most leverage; r_v shifts the score once balance is fixed; ρ_r helps but rarely gates by itself.
- **Landscape:** r_v has the highest weight; Δx improvements don't raise the score if breathing room collapses.
- **MarkMaking-style work:** Δx and ρ_r share the load; centering and under-committed edges both depress K_lite; wild high-frequency chatter can trip ρ_r's upper guard.

Per-set one-liners

- **Landscape (mountains):** pass → narrow fail → pass; *scores track r_v, not Δx*.
- **Abstract (flat graphic):** Δx improves but **r_v goes RED**; last frame still fails on space.
- **Bed (interior):** first passes; recenters and softens → **Δx RED + ρ_r low**.
- **Dancer:** centered → over-open → balanced; only #3 passes (Δx held, r_v contained).
- **Refusal (poster):** centered fail, then **void overwhelms**; needs more subject or tighter field.
- **Sunset:** all **ρ_r-limited**—pretty but airbrushed; wants real edges/texture.
- **Ritual:** all below gate; **Δx is the knob** across the set; color shows real void the gray mask swallows.
- **Farmer:** #0 squeaks by; grid variants read **centered + airless** under gray mask.
- **Figure (atelier):** rides the threshold; tiny r_v dips flip pass/fail; **keep pushing Δx**.
- **Man:** fix Δx once → knob flips to r_v → stable passes.
- **Painter (studio):** three different fails (ρ_r low; Δx+r_v RED; r_v RED); cure = off-center + vent + 1–2 crisp edges.
- **Explosion: ρ_r RED** in every variant—too much shard chatter; compress to a few chosen lines.
- **Hall:** a clean high score proves the meter *can* reward classical order when all gauges cohere
- **Cartoon / Still / Rock / Bridge / Fig crowd:** patterns match profile priors—Bridge and Rock improve as Δx↑ and r_v↑; crowd becomes air-heavy and falls under gate.
- **Skull / Robot / Mount (Cézanne):** Skull under-gate (thin structure); Robot passes until one over-tight variant slips; Cézanne #1/#2 show that iterative painting **can** land in-band when placement/void are chosen, even with brushy ρ_r.

What consistently worked

- **Decisive off-center placement** (lock Δx mid-band, then stop pushing).
- **A real breathing pocket** opposite the weight (vent, sky/water, calmer flank).
- **Two committed edge families** to set ρ_r (not ten thousand micro-edges or pure glaze).

Expected caveats (not bugs)

- Gray mask is **non-semantic**—patterned fields and warm tonals can under/over-credit r_v; color telemetry helps audit this.
- ρ_r is **structure energy**, not brightness contrast; painterly glaze will read low, debris fields high.
- Near the **55 gate**, ±0.5 wiggle from tiny crops/exports is normal.
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Next additions (extend the study)

- Add **optional external subject masks** for a few archetypes (portrait, poster, reflection) as *audit-only* checks.
- Light **ROI helpers** (reflection horizon / sun core) for landscapes to stabilize Δx/ρ_r reads.
- A couple of **genre-tuned profiles** (poster/graphic, still life classic) to show guard/weight movement.
- Report **band distance charts** (normalized) alongside the table in the appendix.

TL;DR

Summary.

Fix **Δx**, contain **r_v**, choose **two edge families**. Color telemetry didn't change any decisions; it clarified *why* gray made the call in tonal, reflective, or patterned scenes.

Concerns (watch-outs).

Mask semantics on patterned grounds; painterly glaze reading low ρ_r; pushing Δx so far the knob flips to r_v; living on the 55 line.

Opportunities (use it to win).

- Use **priority knob + direction-to-center** as the one-move brief for the next iteration.
- Use **color telemetry** to explain divergences to stakeholders without reopening the gate.

- Treat fails as **within-band rebalancing**, not rejection of taste.

Real-world playbook (do this next).

1. Lock Δx in-band; 2) open or contain r_v per profile; 3) add two decisive edge families; 4) re-run; 5) when gray vs color disagree, screenshot the color audit—don't change gating.

Where it already pays off (from the sets).
Man, Bridge, Rock, Figure, Landscape (mountains) all became *predictably better* once the knob was followed; Sunset and Painter show exactly *what* edge work earns passage; Explosion shows what to **remove** (high-freq chatter).

Bottom line.
The lite meter is doing its job: **name the lever, point the direction, reward discipline**. Color telemetry is the right kind of "more"—it explains, it doesn't argue.